

AHSANULLAH UNIVERSITY OF SCIENCE AND TECHNOLOGY
DEPARTMENT OF MECHANICAL AND PRODUCTION ENGINEERING



FINAL REPORT

COURSE NO : IPE 3112

COURSE TITLE : PRODUCT DESIGN AND DEVELOPMENT SESSIONAL

PRODUCT NAME : ERGONOMIC CUTTING TOOL

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Chapter 01

Introduction

In product design, we recognize that the integration of functionality, user comfort, and cost-effectiveness has become essential for developing competitive and practical solutions. Cutting tools are widely used in households, workshops, and light industrial environments; however, many conventional designs primarily focus on functionality while often overlooking ergonomic considerations. As a result, prolonged use of such tools can lead to hand fatigue, discomfort, reduced control, and decreased overall efficiency.

In This project we aimed to design and develop an Ergonomic Cutting Tool that addresses these limitations by incorporating user-centered design principles. The objective is to enhance user comfort, improve grip stability, and ensure efficient cutting performance while maintaining durability and economic feasibility. To achieve this, a systematic product design and development approach has been followed.

Our study began with a Market Analysis to identify target users and understand their needs and expectations. Based on these insights, Quality Function Deployment (QFD) is applied to translate customer requirements into measurable engineering characteristics. Subsequently, Functional Structure Development (FSD) is carried out to analyse and represent the functional relationships within the product.

During the design phase we performed detailed design and specification analysis, supported by modelling and evaluation of key parameters such as strength, usability, and performance. A Design of Experiments (DOE) approach is then utilized to optimize critical design factors and determine the most effective combination of parameters. Furthermore, appropriate Material Selection is performed considering strength, durability, corrosion resistance, and cost.

Finally, a Cost Analysis is conducted to evaluate the economic feasibility of the proposed design, ensuring that the product remains viable for practical implementation and potential mass production. The report concludes with a summary of findings, along with identified limitations and possible directions for future improvement.

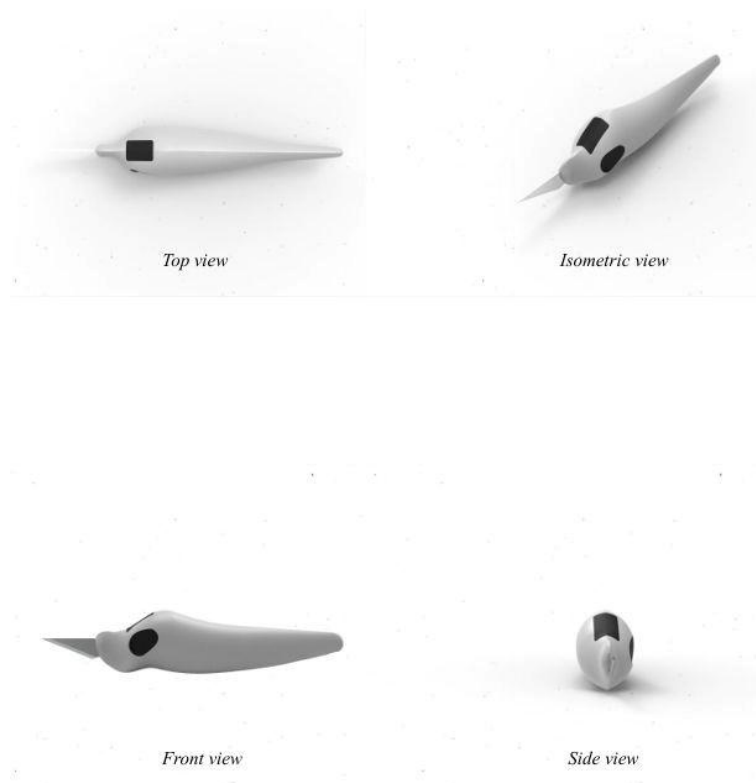
Chapter 02

Market Analysis

PRODUCT NAME

Ergonomic Cutting Tool

PRODUCT IMAGE



INTRODUCTION

Market analysis is a systematic study of existing market conditions to understand customer needs, product demand, competitor offerings, and potential opportunities for improvement. It helps in identifying gaps between currently available products and user expectations. In product design and development, market analysis plays a crucial role in ensuring that a proposed product is practical, relevant, and commercially viable.

For the Ergonomic Cutter, market analysis is conducted to evaluate existing cutting tools, identify their limitations, and assess user requirements, particularly in garments and textile-

related applications. The findings from this analysis provide a foundation for design decisions and ensure that the proposed product addresses real world problems effectively.

NEED STATEMENT

Cutting tools are widely used in garments, textile, packaging, and manufacturing industries for repetitive cutting operations. Most conventional cutters available in the market are designed with straight handles and uniform grip surfaces, which often lead to awkward wrist posture and concentrated pressure on the hand during prolonged use. As a result, users may experience hand fatigue, discomfort, and reduced efficiency over extended working periods.

There is a need for an ergonomic cutter that addresses these issues by improving handle geometry and grip comfort without increasing complexity or cost. An ergonomically designed cutter can help maintain a natural wrist posture, distribute applied force more evenly across the palm, and enhance grip stability. This is particularly important in garments and textile industries, where workers perform continuous cutting tasks for long durations. The proposed Ergonomic Cutter aims to fulfill this need by focusing on user comfort, safety, and productivity through passive ergonomic design improvements.

TARGET CUSTOMER

The target customers for the ergonomic cutter are individuals and organizations that frequently use cutting tools in their daily tasks and are exposed to hand strain, fatigue, or risk of injury due to prolonged or repetitive cutting activities. This product is designed especially for users who prioritize comfort, safety, and efficiency in manual operations. By reducing wrist stress and improving grip control, the ergonomic cutter addresses both productivity and occupational health concerns, making it suitable for professional as well as regular users.

Target customer groups include:

- Industrial and factory workers involved in packaging, trimming, or material cutting
- Warehouse and logistics personnel handling cartons and packaging materials
- Office employees who frequently use cutters for stationery and document-related tasks

- Craftsmen, DIY enthusiasts, and hobbyists requiring precision cutting
- Educational institutions and students for academic and project based use
- Individuals with hand discomfort, repetitive strain injury (RSI), or reduced grip strength

AREAS AND LOCATION OF SURVEY

The market survey was conducted using an online Google Form to collect responses from a diverse group of users. The focus was on both academic and general users to understand different usage needs and preferences related to an ergonomic cutter.

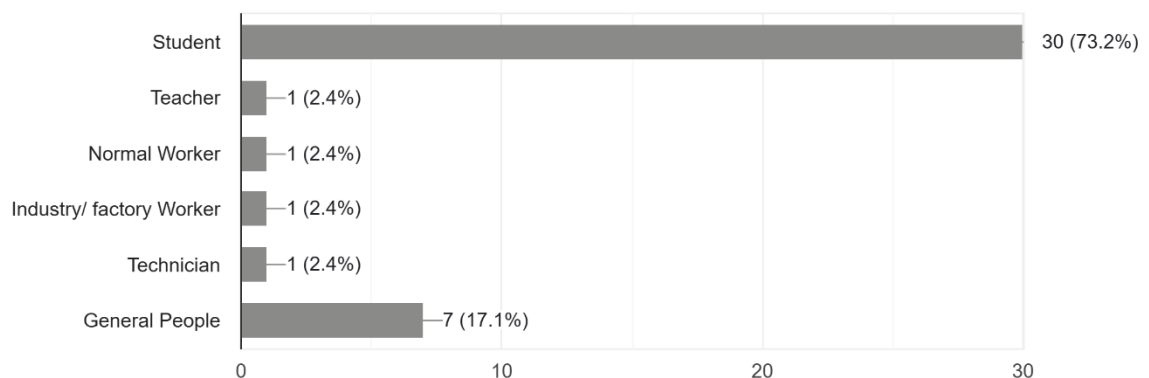
- University students from various departments and academic years
- General public including office workers, shopkeepers, and household users
- Respondents from urban and semi-urban areas
- Survey conducted primarily through online platforms for ease of access and wider reach

SURVEY RESULTS

Question 01

Tell us a bit about yourself-

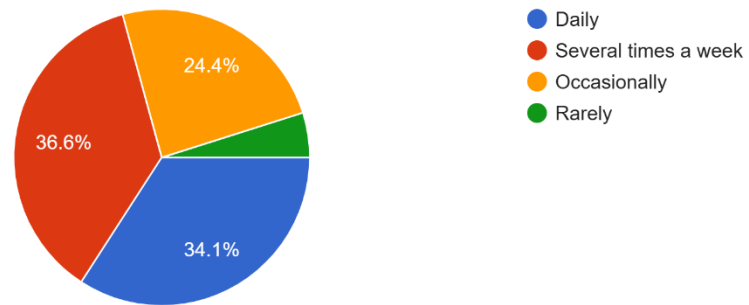
41 responses



Question 02

How often do you use a cutter or cutting tool?

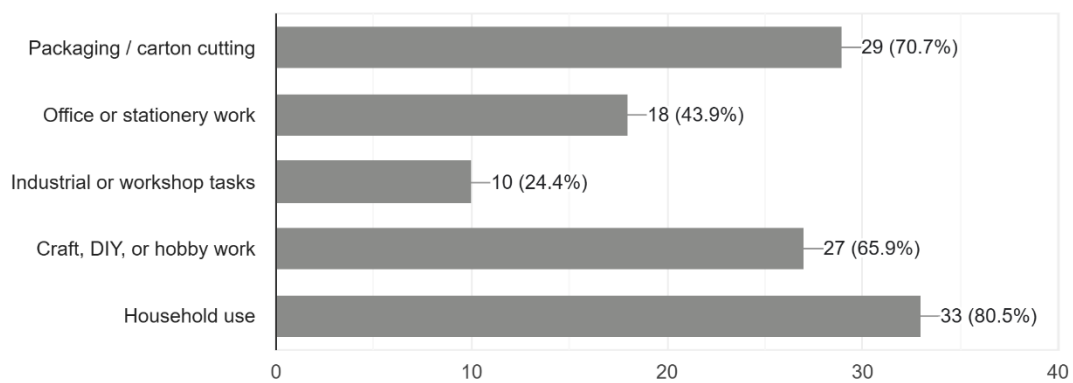
41 responses



Question 03

For what purposes do you mostly use a cutter?

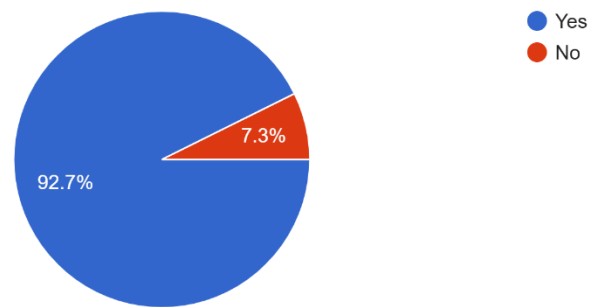
41 responses



Question 04

Have you ever accidentally cut yourself while using a conventional cutter?

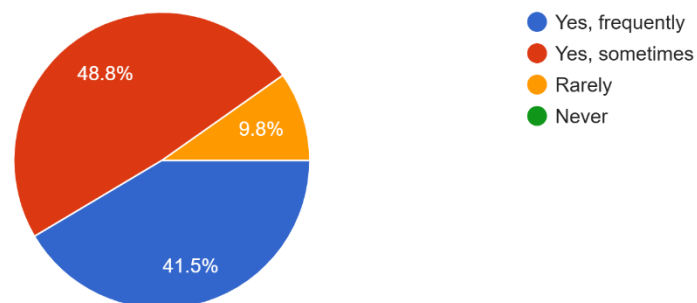
41 responses



Question 05

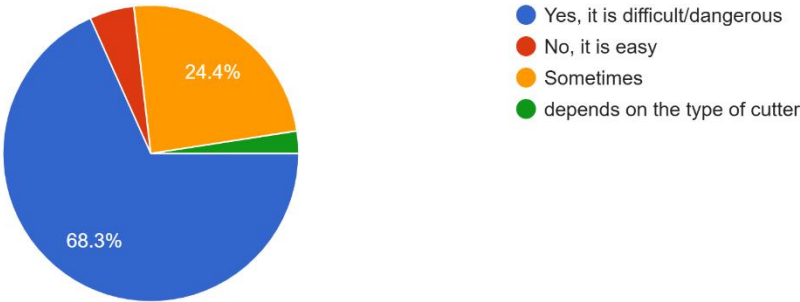
Have you ever experienced hand fatigue or discomfort while using a cutter?

41 responses



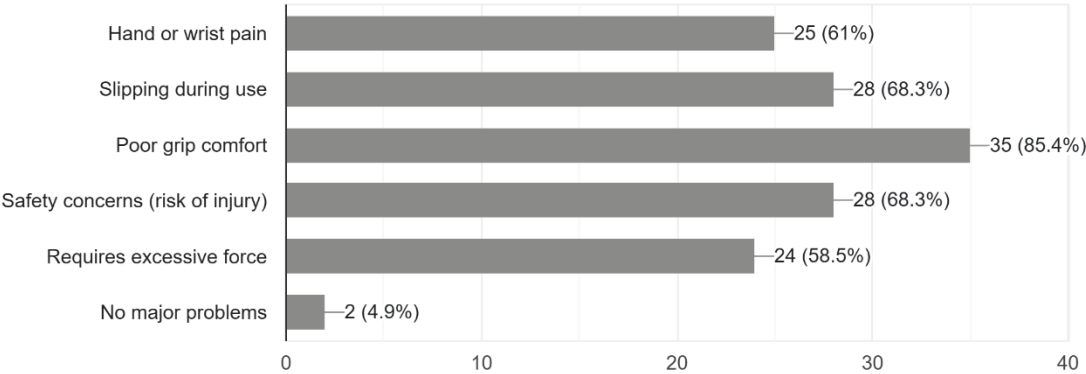
Question 06

Do you struggle with changing the blades on your current cutter?
41 responses



Question 07

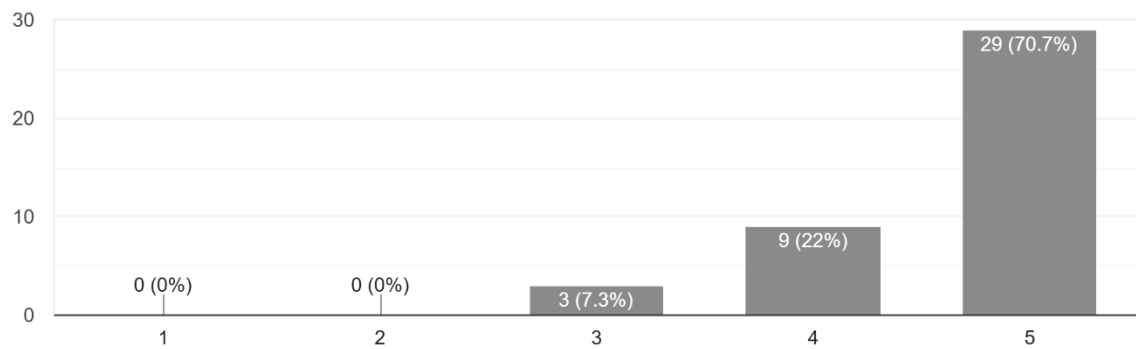
Which problems do you usually face with conventional cutters?
41 responses



Question 08

How important is ergonomic design (comfortable grip and reduced strain) in a hand tool?

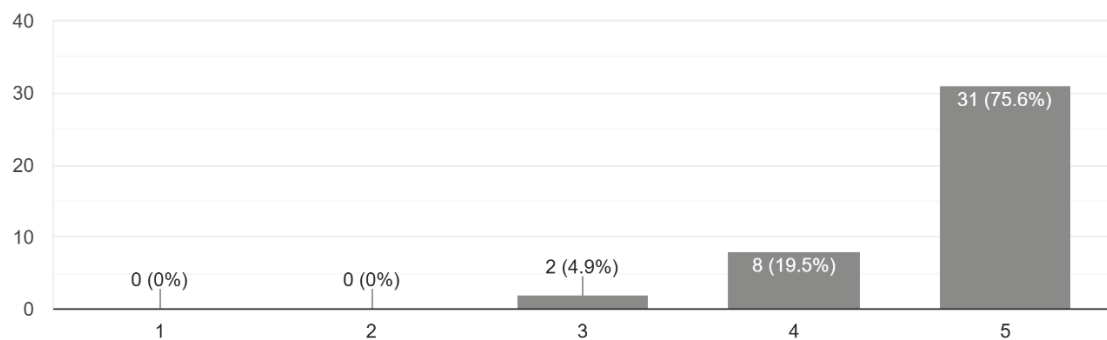
41 responses



Question 09

If a cutter is designed to reduce hand strain and improve comfort, how likely are you to use it?

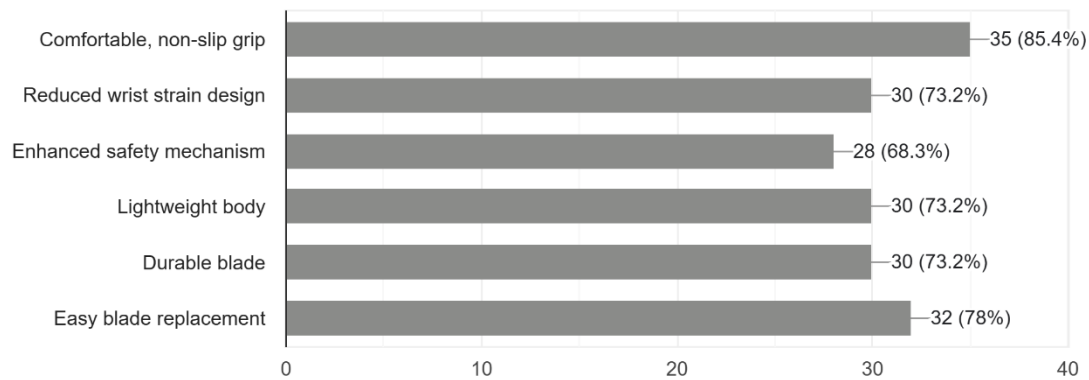
41 responses



Question 10

Which features would you prefer in an ergonomic cutter?

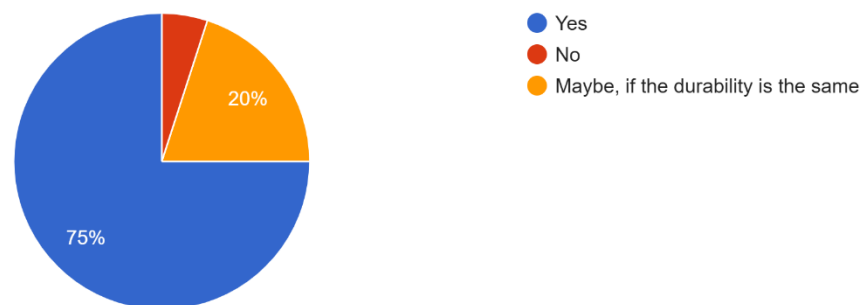
41 responses



Question 11

Would you be interested in a cutter made from recycled or eco-friendly materials?

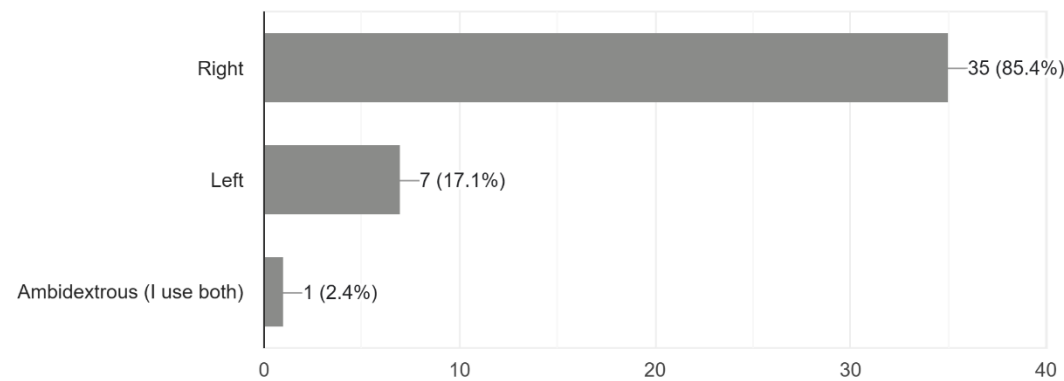
40 responses



Question 12

Which hand is your dominant hand? (Critical for ergonomic design!)

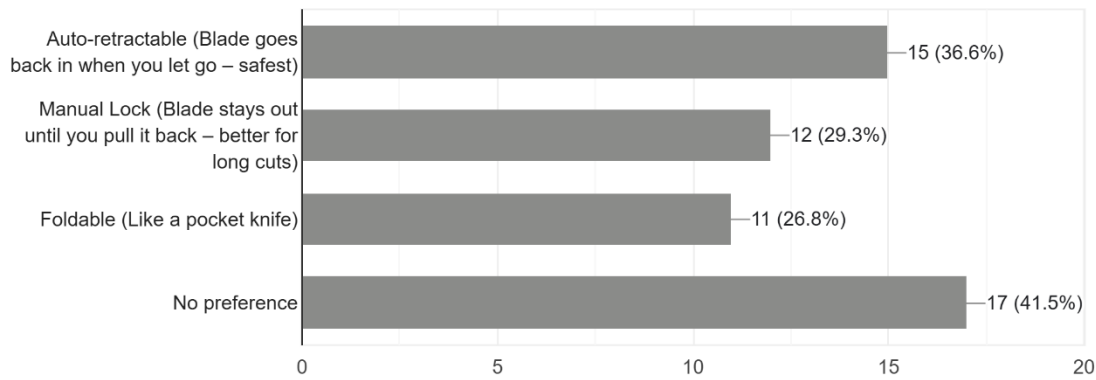
41 responses



Question 13

Which blade mechanism do you prefer?

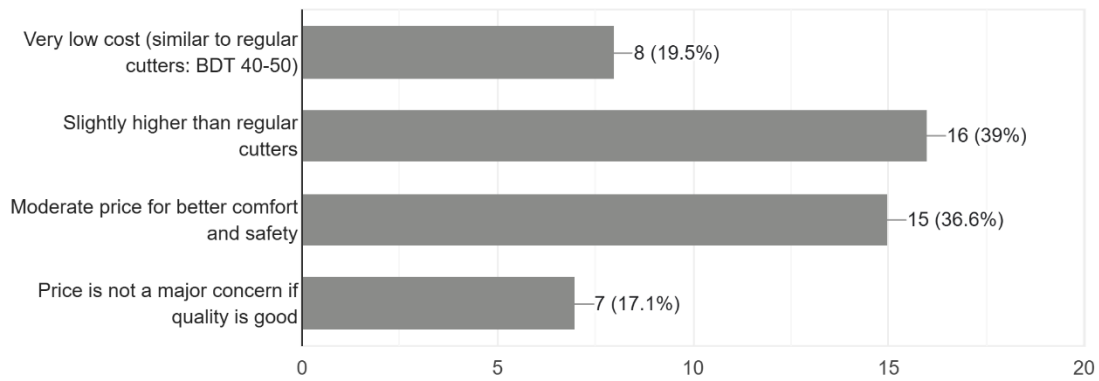
41 responses



Question 14

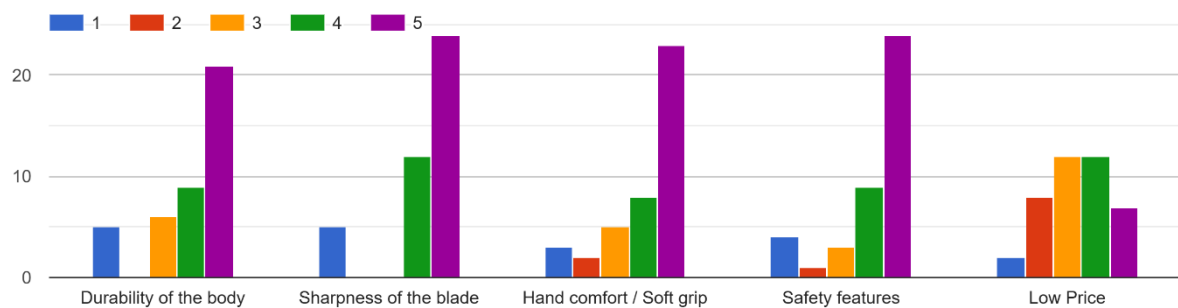
What price range would you consider reasonable for an ergonomic cutter?

41 responses



Question 15

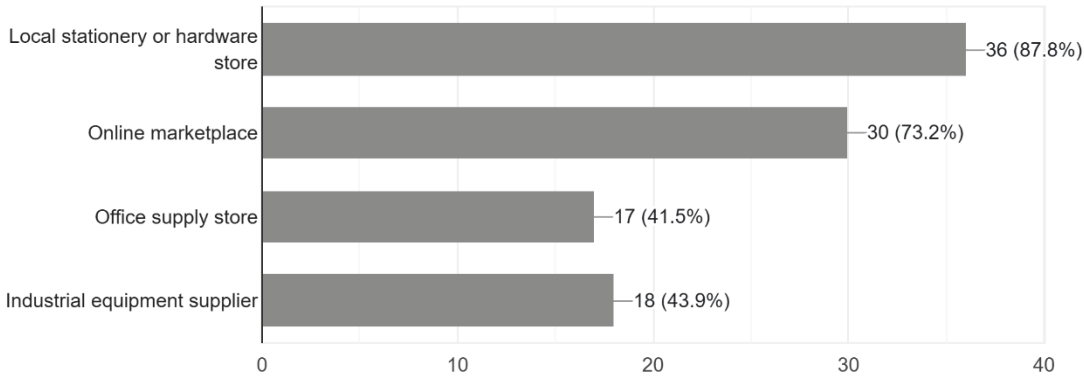
Please rank the following features in order of importance (1 = Most Important, 5 = Least Important):



Question 16

Where would you prefer to purchase an ergonomic cutter?

41 responses



Question 17

What additional features or improvements would you like to see in an ergonomic cutter?

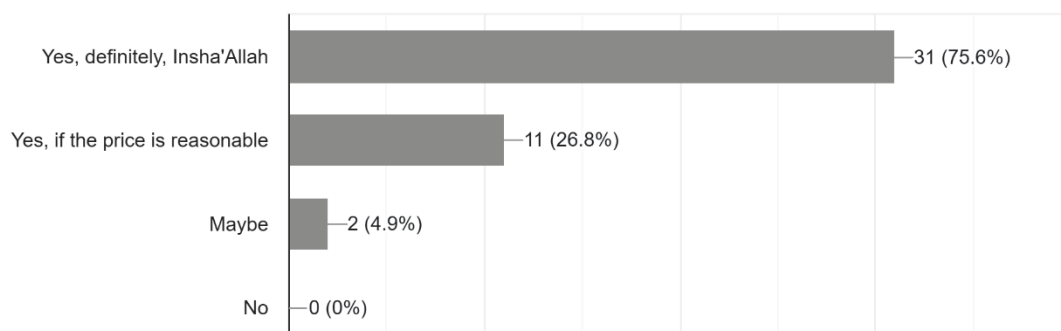
10 responses

N/A
Very good
A smart storage case that also works as a blade disposal unit to prevent accidental injuries
Don't have any
A visual blade-wear indicator that shows when the blade needs replacement to maintain cutting quality and safety.
Please add a miniature version so that i can carry it around.
Color-coded variants based on application (office, textile, industrial) to improve identification and reduce misuse
A detachable mini ruler or measurement markings on the body for quick and accurate cutting without needing a separate scale.

Question 18

Overall, would you be willing to switch from a conventional cutter to an ergonomic cutter if we make such a cutter?

41 responses



KEY FINDINGS

Respondent Profile

The survey responses were collected from a diverse group of participants, with the majority being university students, followed by general users from various professional and household backgrounds. This mix ensured that the findings reflect both academic and real life usage of cutting tools, making the analysis more representative of the target market.

Usage Frequency and Application Areas

The results showed that cutters are frequently used tools, with most respondents using them daily or several times a week. The primary application areas included household use, packaging and carton cutting, office and stationery work, as well as industrial and workshop tasks. This indicates that cutters are essential tools across multiple sectors.

Problems with Conventional Cutters

A significant number of respondents reported experiencing accidental cuts, hand fatigue, or discomfort while using conventional cutters. The most common issues identified were hand and wrist pain, slipping during use, poor grip comfort, safety concerns, and the need to apply excessive force. These findings highlight clear shortcomings in existing cutter designs.

Importance of Ergonomic Design

Most respondents rated ergonomic design such as comfortable grip and reduced hand strain as highly important in a hand tool. This demonstrates a strong awareness among users regarding comfort and safety, especially for tools used repeatedly over long periods.

User Interest and Willingness to Switch

Most respondents expressed a high level of interest in using an ergonomic cutter and indicated a strong willingness to switch from conventional cutters, particularly if the product is priced reasonably. This reflects positive market acceptance and readiness for an improved cutting tool.

Preferred Features

The survey revealed that users prefer features such as comfortable grip design, reduced wrist strain, enhanced safety mechanisms, lightweight construction, durable blades, and easy blade replacement. These preferences directly support the proposed design focus of the ergonomic cutter.

Purchase Preference

When asked about purchasing locations, most respondents preferred buying an ergonomic cutter from local stationery and hardware stores. This suggests that easy accessibility and physical availability are important factors influencing purchasing decisions.

Suggested Improvements and Value Addition

Respondents also suggested additional value adding features, including eco-friendly or recyclable materials, blade wear indicators, built-in measurement markings, safe blade storage or disposal options, and modular or color coded designs. These suggestions indicate opportunities for product differentiation in the market.

Overall Market Potential

Overall, the survey results confirm strong market potential for the ergonomic cutter. The findings clearly indicate a demand for a safer, more comfortable, and user-friendly cutting tool, and demonstrate that users are willing to adopt an improved design that effectively addresses the limitations of conventional cutters.

CONCLUSION

The market analysis conducted through the survey provides strong support for the development of the proposed ergonomic cutter. The findings indicate that cutting tools are widely used across academic, household, office, and industrial environments, and that users frequently experience discomfort, safety issues, and fatigue while using conventional cutters. These problems highlight the need for improvement in existing cutter designs.

The survey results show that ergonomic considerations such as comfortable grip, reduced hand strain, and improved safety are highly valued by users. A significant majority of respondents expressed willingness to switch to an ergonomic cutter, especially if it is offered at a reasonable price and made easily available through local stationery and hardware stores. Additionally, user feedback suggested several value adding features, indicating opportunities for product differentiation and enhanced user satisfaction.

Overall, the market analysis confirms that there are a clear demand and positive acceptance for an ergonomic cutter. The insights obtained from the survey provide a strong foundation for the design and development phase, ensuring that the proposed product is user-centered, practical, and aligned with real market needs.

Chapter 03

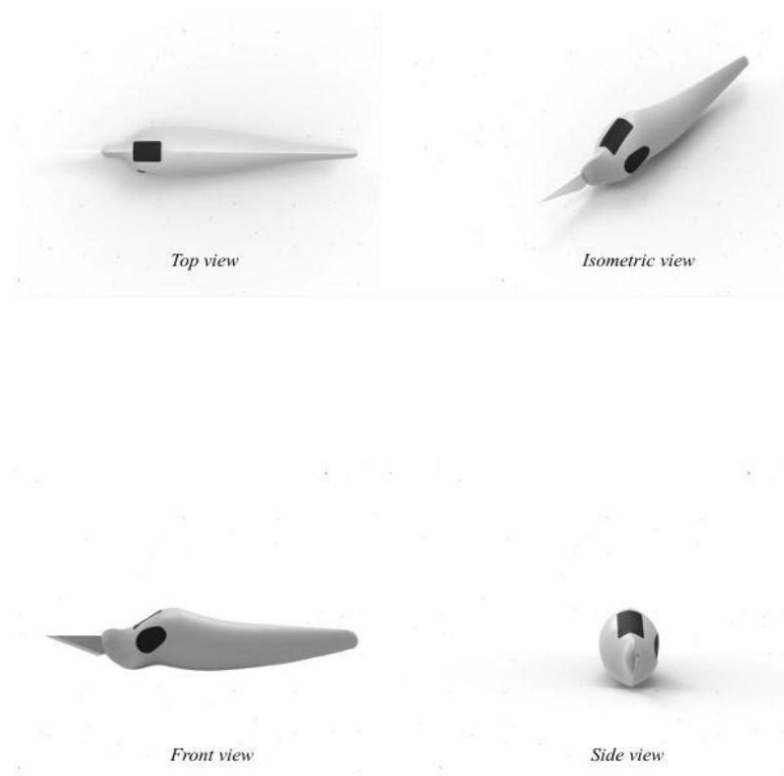
Quality Function Deployment

3.1 INTRODUCTION

Quality Function Deployment (QFD) is a structured product development methodology used to translate customer needs and expectations into specific technical and engineering characteristics. The primary objective of QFD is to ensure that the voice of the customer is systematically incorporated into the design and development process, thereby improving product quality, customer satisfaction, and overall design efficiency. By identifying customer requirements early and linking them directly with design parameters, QFD helps reduce redesign efforts, development time, and costs.

The House of Quality (HOQ) is the core tool of QFD and serves as a matrix-based framework to establish relationships between customer requirements (WHATs) and technical characteristics (HOWs). It visually represents how well engineering attributes address customer needs and highlights priorities, correlations, and trade-offs among technical parameters. Through the HOQ, organizations can make informed decisions, set design targets, and focus resources on the most critical aspects of product development, ensuring a customer-driven and systematic approach to quality improvement.

3.2 PRODUCT FIGURE



3.3 Framework of a house:

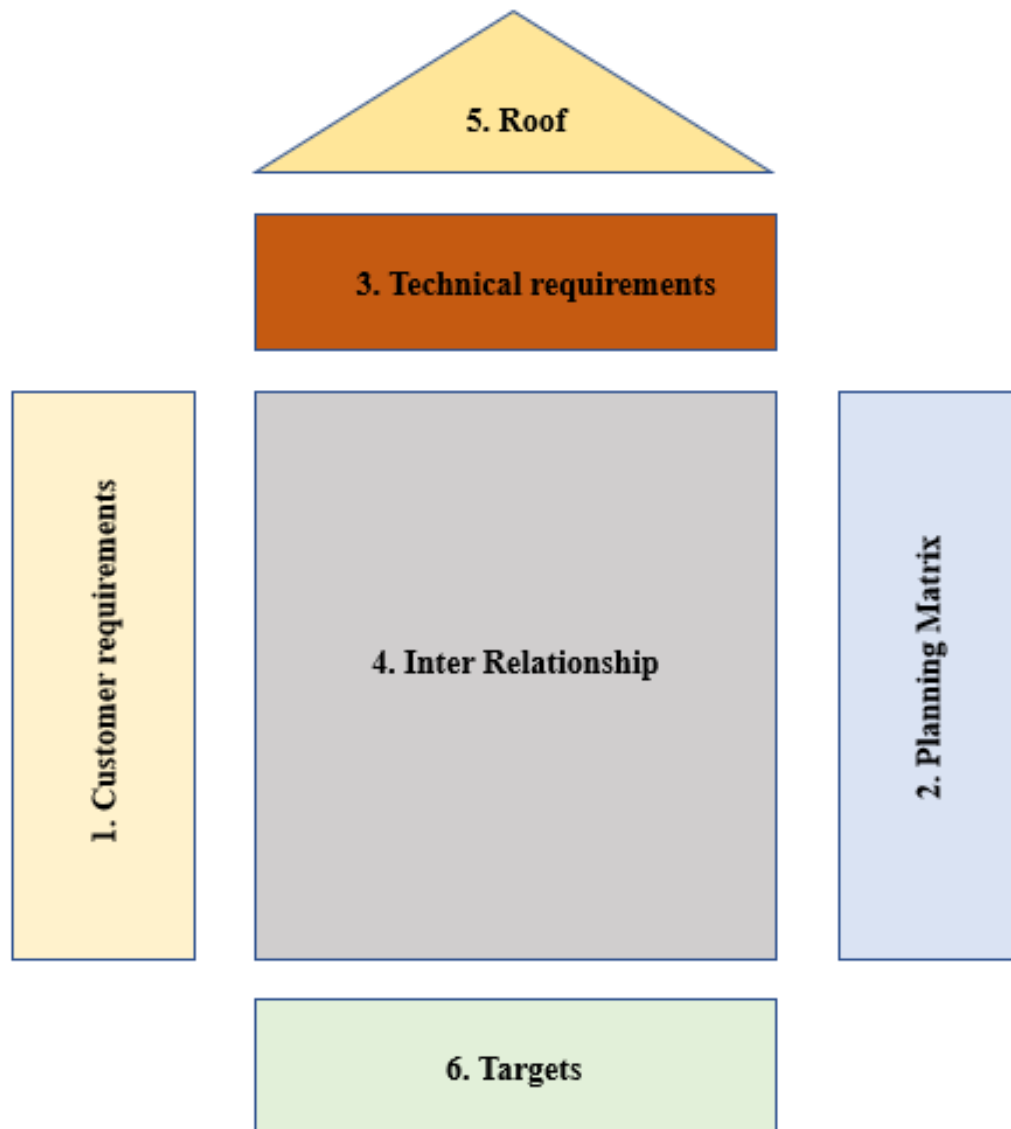


Figure 3.3.1: Framework of a house

3.3.1 Table for relationship matrix:

Relationship Matrix	
Strong	9
Medium	3
Weak	1
No relation	0

3.3.2 Correlation matrix:

Correlation Matrix	
++	Strong Positive
+	Positive
-	Negative
--	Strong Negative
	Not correlated

3.3.3 Table for relationship matrix:

Competitor Analysis
P= Poor
F= Fair
G= Good

3.3.4 Table for relationship matrix:

Customer rating	
Strong relationship	Rating 5
Moderate relationship	Rating 3
Weak relationship	Rating 1

3.3.5 House of Quality

Technical Requirements Customer Requirements												
--	--	---	--	--	--	--	--	--	--	--	--	--

3. 4 Relationship between Technical Requirements

We have analyzed the relationship between the technical requirements using the correlation matrix in the House of Quality. This analysis helps us understand how one technical parameter affects another during the design process.

We have found that some technical requirements have a positive relationship, meaning that improving one requirement also improves another. For example, when we improve blade geometry, the cutting force is reduced, which improves the overall cutting performance. On the other hand, we have also observed negative relationships between certain technical requirements. For instance, increasing the mass and balance of the product can improve stability, but it may increase user fatigue during repetitive use.

By identifying these positive and negative relationships, we have been able to recognize possible design trade-offs. This has helped us make balanced design decisions so that improvement in one technical area does not negatively affect the overall performance of the product.

3. 5 Relationship between Technical and Customer Requirements

We have established the relationship between customer requirements and technical requirements using the relationship matrix. In this matrix, we have assigned values such as strong (9), medium (3), weak (1), and no relationship (0) to show how effectively each technical requirement meets customer needs.

From our analysis, we have found that customer requirements such as safety, low cutting force, fine control, and comfortable grip are strongly related to technical requirements like blade exposure, guard/lock reliability, blade geometry, and grip hardness. This indicates that these technical features are most important for satisfying customer expectations.

By combining the customer importance ratings with the relationship values, we have calculated the technical importance ratings. Based on these results, we have identified cutting force, blade geometry, and blade exposure as the most critical technical requirements.

Through this method, we have successfully translated customer needs into specific technical parameters, which will help ensure that the final product design meets customer expectations effectively.

3. 6 Conclusion:

We constructed the House of Quality (HOQ) as the main tool in Quality Function Deployment (QFD) by first collecting the voice of the customer (VOC) to understand their needs. These customer requirements were then translated into technical specifications. We analysed the relationships between customer needs and technical requirements, assessed correlations among technical features, compared competitors, and set measurable targets. This systematic approach allowed us to prioritize important features, guide product design, and ensure higher customer satisfaction while minimizing redesign.

Chapter 04

Functional Structure Development

4.1 INTRODUCTION

Functional structure development is a systematic method used in product design to describe what a product must do in terms of functions and flows of material, energy, and information. The process usually begins with a black box representation, where only the overall function of the system and its inputs and outputs are shown, without specifying any internal components or mechanisms. From this overall view, a function tree is created by breaking the main function into smaller, logically ordered subfunctions that represent the individual actions needed to achieve the desired output.

These subfunctions are then arranged into a functional structure, where each function is linked to corresponding flows of material, energy, or information, giving a detailed picture of how the system transforms inputs into outputs. Reverse engineering can be used alongside this process by analysing existing products, identifying their main functions and subsystems, and expressing them in the form of a function tree and functional structure to reveal strengths, weaknesses, and opportunities for improvement. In this way, functional structure development provides a solution-neutral foundation that supports the later design and refinement of products such as an Ergonomic Cutting Tool.

4.2 BLACK BOX DIAGRAM



Fig: Black Box Diagram for Ergonomic Cutting Tool

This black box diagram represents the overall function of the Ergonomic Cutting Tool by showing only its inputs and outputs in a simplified form. Hand force and motion, the material to cut, and user control or intent enter the system as inputs, while reduced hand strain, the cutting of material, and safe cutting action leave the system as outputs, indicating how the tool transforms user effort and material into an efficient and safer cutting result.

4.3 Functional Tree:

The Function Tree is a structural diagram used to break down the primary objective of the ergonomic cutting tool into manageable, interconnected sub-functions. By treating the device as a "black box," this method logically organizes the operational sequence—from the initial user input (grip and force application) to the final output (material separation).

This hierarchical approach ensures that every mechanical step is executed in the correct order. By mapping these dependencies, the function tree provides a clear visual pathway of the tool's operation, facilitating a design that maximizes both cutting efficiency and ergonomic comfort.

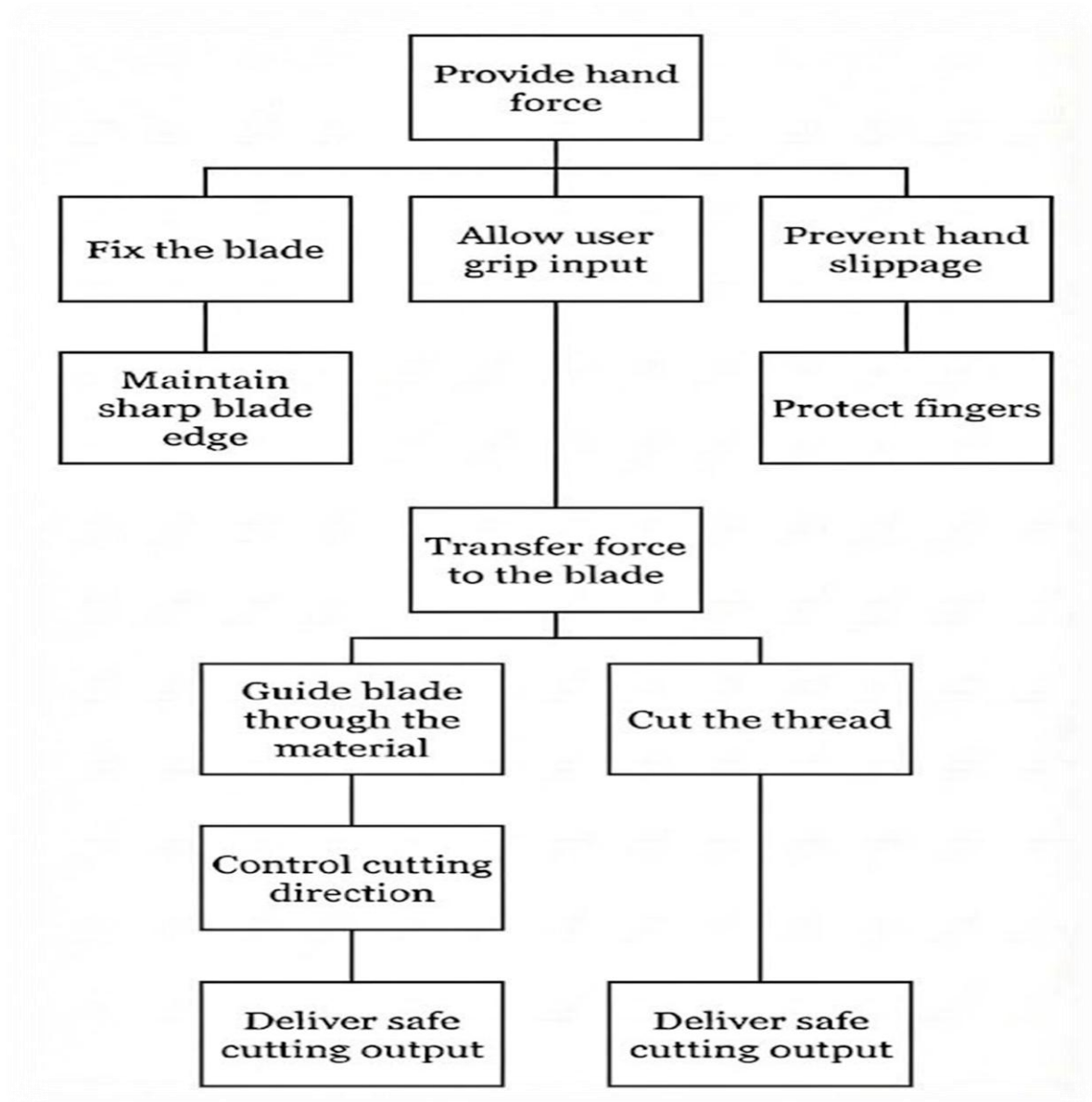


Figure.4.3.1: Functional Tree diagram for Ergonomic Cutting Tool

4.4 Functional Structure:

Functional structure systematically decomposes a product's overall function into a detailed hierarchy of sub-functions through a process called functional decomposition, visually represented as a function tree within a boundary box that defines inputs, outputs, and interactions. We can break down step-by-step to show how raw inputs like energy, materials,

or signals are transformed through internal mechanisms into desired outputs. By mapping this flow clearly, designers gain insights into each component's role, enabling exploration of alternative solutions, identification of inefficiencies, and targeted improvements to enhance performance, reliability, and manufacturability. Commonly used in engineering fields like drone design or mechanical systems, it fosters innovative problem-solving without physical prototypes, bridging abstract requirements to concrete implementations.

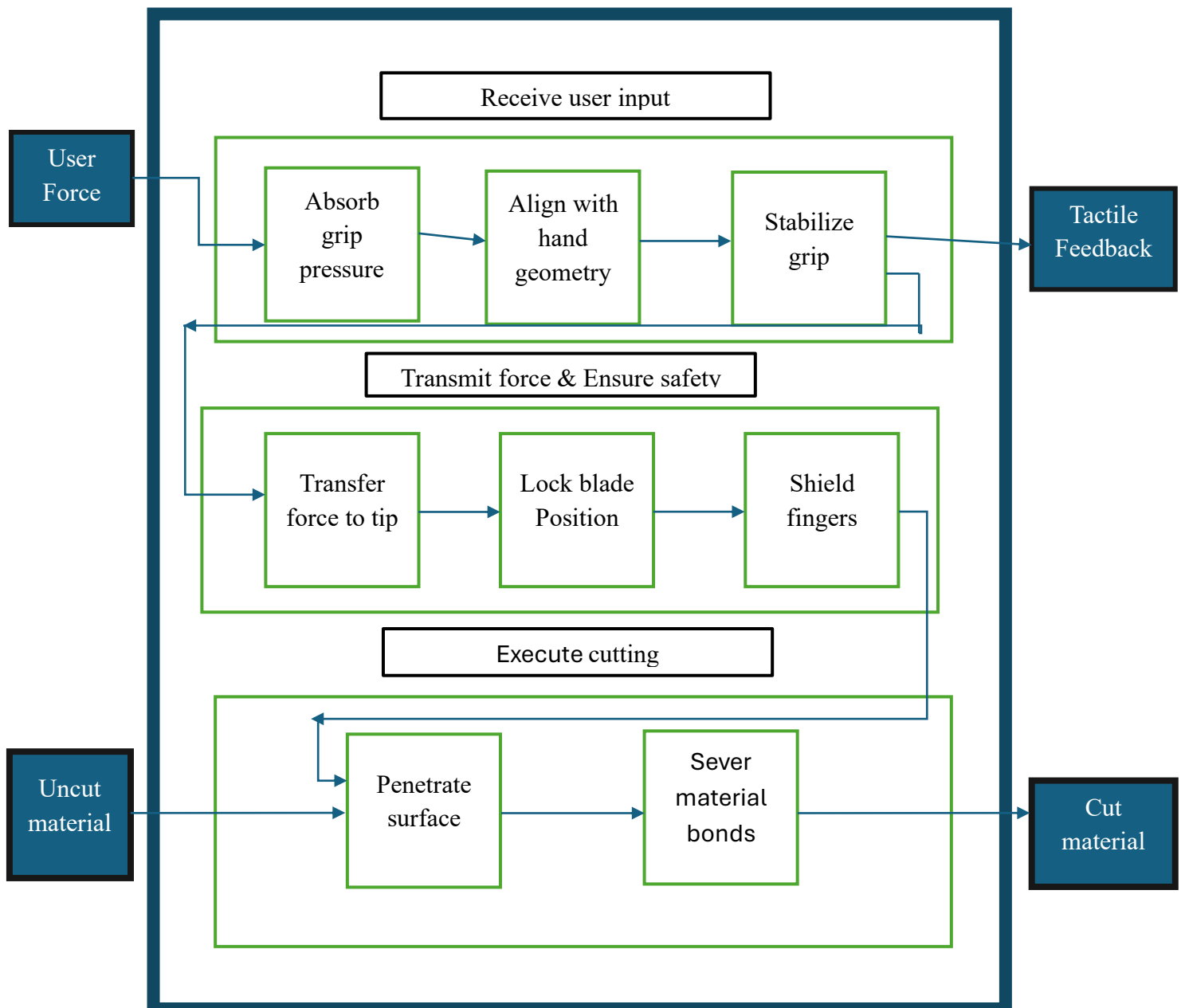


Figure.4.4.1: Functional Structure diagram for Ergonomic Cutting Tool

4.5 Reverse Engineering:

Reverse engineering systematically dissects an existing product or system into its core subsystems to uncover design principles, functionality, and performance characteristics without relying on original documentation. Engineers begin by disassembling components, measuring geometries with tools like 3D scanners or callipers, and analysing materials, tolerances, and interactions to build a solution-neutral function tree that maps out how the product achieves its goals. This breakdown reveals strengths, weaknesses, and inefficiencies, enabling targeted improvements such as integrating advanced materials, optimizing manufacturing processes, or enhancing interoperability with modern technologies. Widely applied in industries like automotive, aerospace, and consumer electronics, it supports competitive analysis, recreation of obsolete parts, and rapid prototyping through CAD modelling and simulations. Ultimately, the process fosters innovation by translating empirical insights into refined specifications for superior next-generation designs.

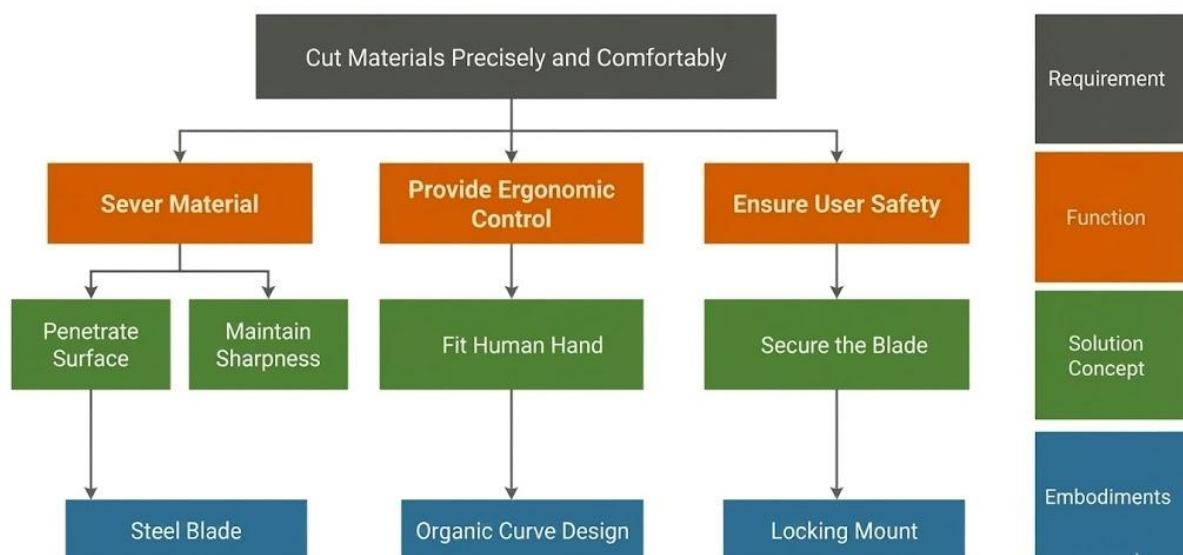


Figure.4.5.1: Reverse Engineering diagram for Ergonomic Cutting Tool

4.6 Conclusion:

Through functional decomposition, the ergonomic cutting tool emerges as a cohesive system designed to "cut materials safely, efficiently, and with minimal user strain," where the primary function decomposes into essential sub-functions including secure hand gripping, ergonomic force application via levers, precise blade alignment and shearing, and controlled motion feedback to prevent slippage or fatigue. Represented through black box diagrams and detailed function trees, this structure illustrates how inputs like user hand force, material to be cut, and environmental signals transform into outputs such as clean incisions and user comfort, highlighting the interplay of energy flows (muscular effort), material flows (blade-material contact), and signal flows (tactile feedback). Reverse engineering further refines this by pinpointing legacy weaknesses in traditional cutting tools such as poor pivot efficiency or rigid handles and enabling upgrades with advanced grips, spring-assisted mechanisms, or adjustable ergonomics tailored for prolonged use in professional or home settings. In essence, this comprehensive functional mapping not only validates the tool's superior design principles but also guides prototyping, manufacturing optimizations, and iterative improvements, ensuring the final product excels in precision, durability, safety, and user satisfaction across diverse applications.

Chapter 05

Specification and Design Analysis

5.1 DESIGN ANALYSIS

This chapter presents the design analysis of the proposed ergonomic cutting tool with the aim of evaluating its structural strength, safety, and performance under operational loading conditions. The tool has been designed by considering ergonomic principles to reduce user fatigue while ensuring effective cutting performance. Computer-aided design (CAD) modelling has been used to develop the geometry of the tool and its main components.

Finite element analysis (FEA) has been carried out using SolidWorks Simulation to analyze

stress, strain, deformation, and factor of safety of the critical components of the ergonomic cutting tool. The applied loads and boundary conditions represent realistic working scenarios. The results obtained from the analysis are used to verify that the stresses remain within allowable material limits, ensuring the safety, durability, and reliability of the tool before manufacturing.

5.2 MAIN COMPONENTS OF THE PRODUCT

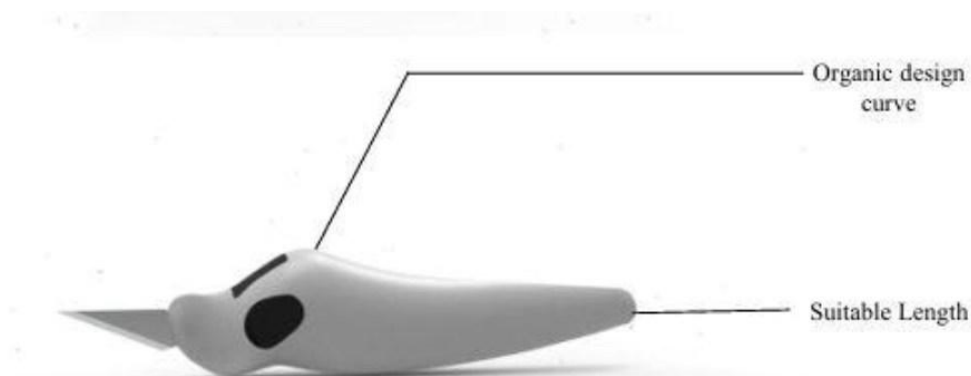


Figure: Front View

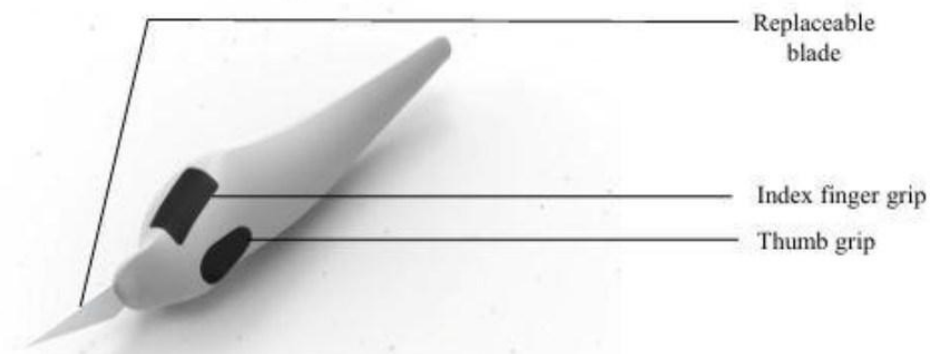


Figure: Isometric View

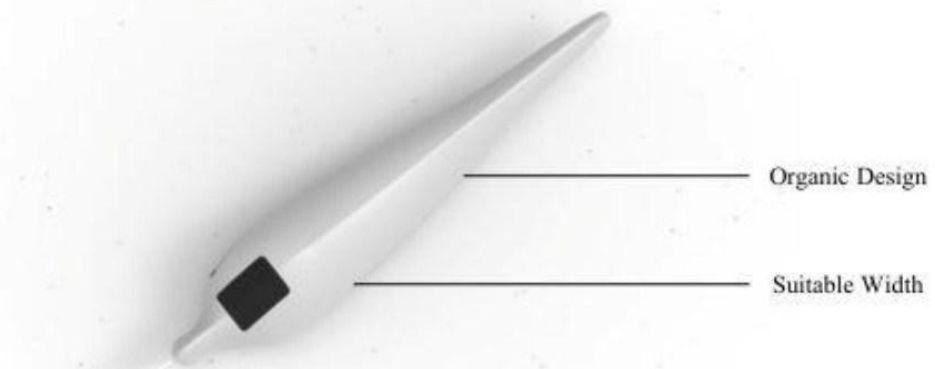


Figure: Top View

5.3: Main Product View

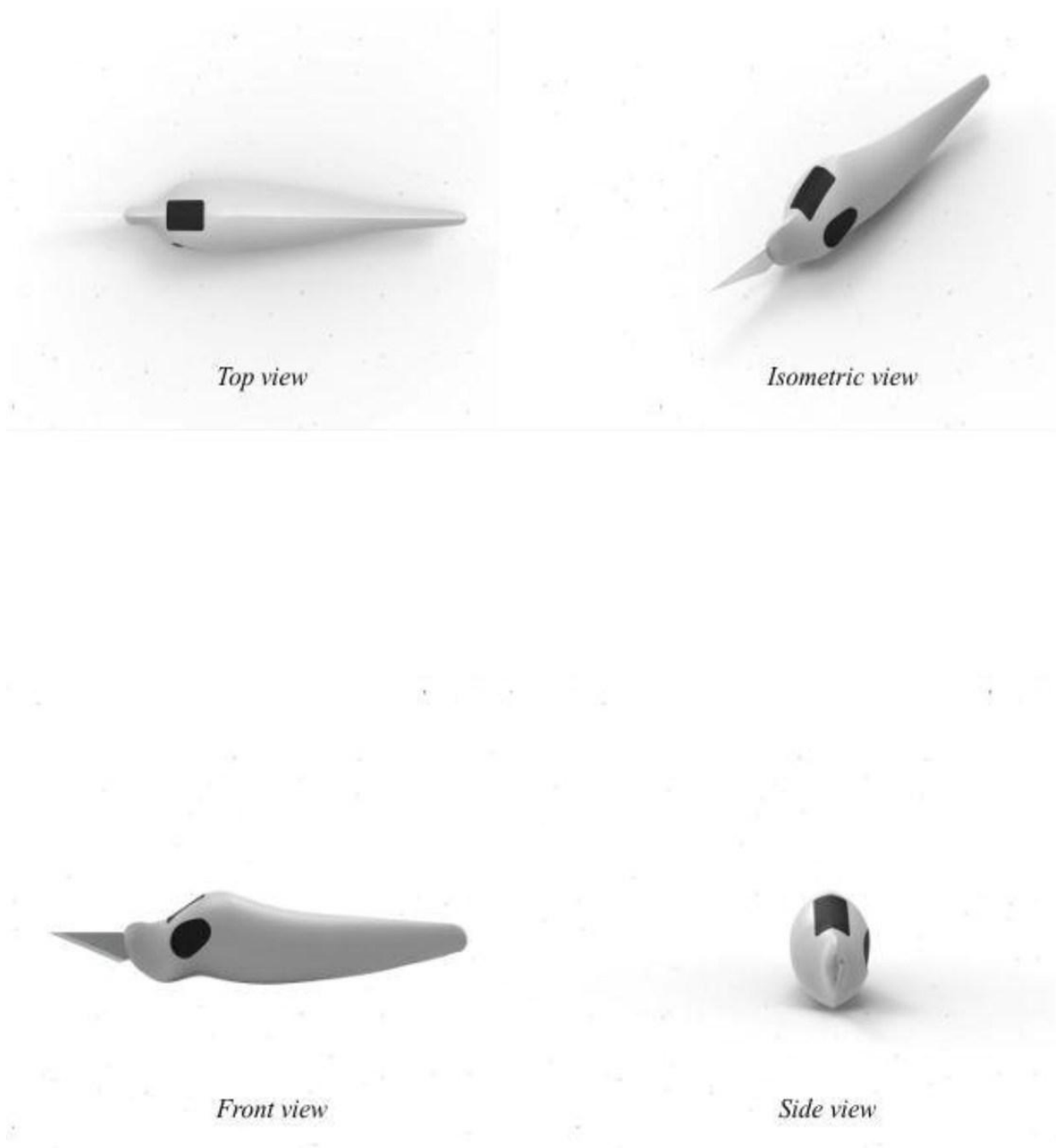


Fig: Different Views of Ergonomic Cutting Tool

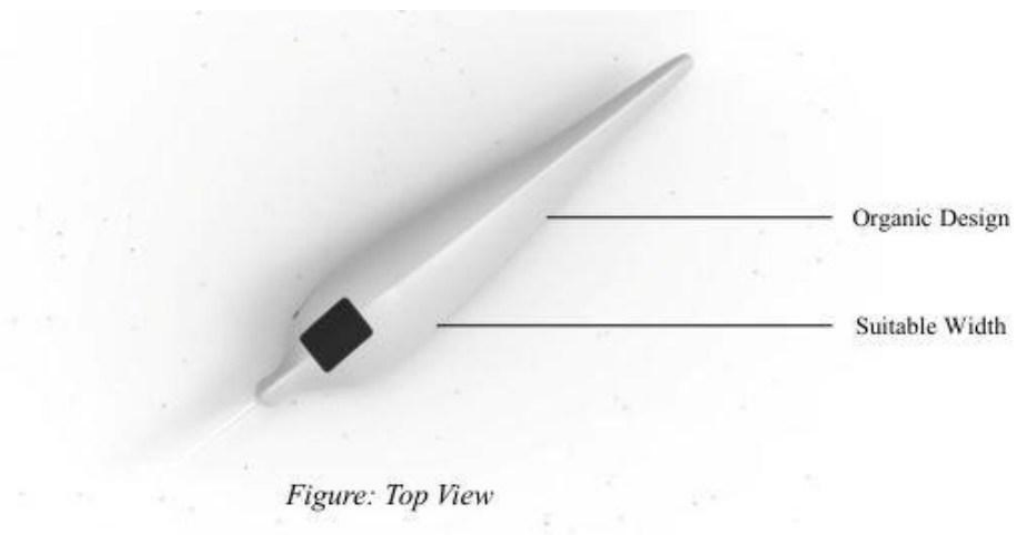
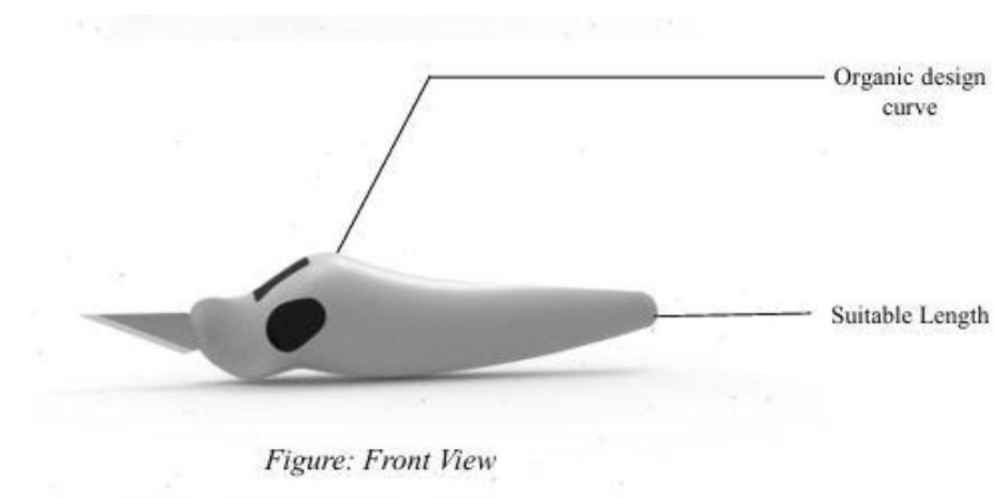
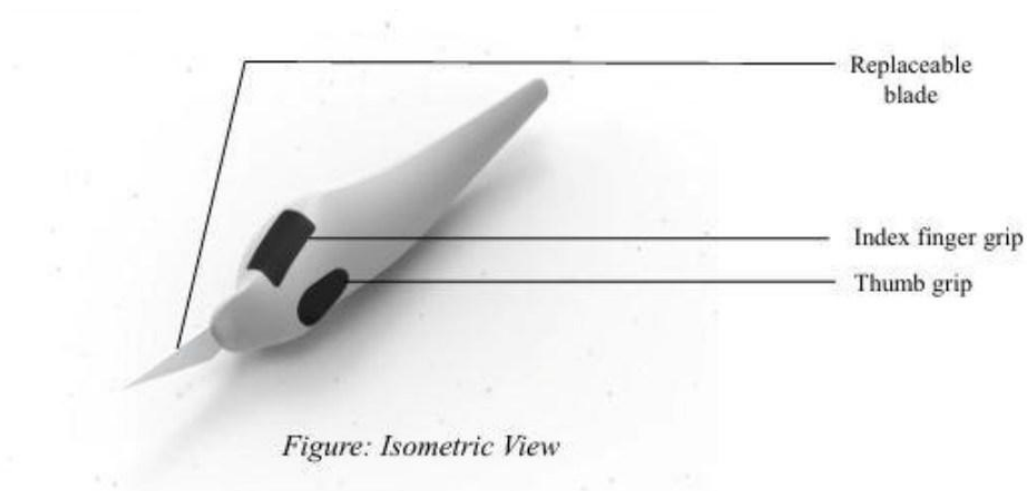


Fig: Individual Views of Ergonomic Cutting Tool



Fig: Different Components of Ergonomic Cutting Tool

5.4: Main component Stress, Deformation, Strain Analysis and Factor of Safety Analysis:

Stress analysis is used to evaluate whether the ergonomic cutting tool can safely withstand the forces applied during normal operation without permanent deformation or failure. The analysis helps identify critical stress concentration zones (such as the blade-seat region, neck/transition area between handle and head, and the trigger/slider support) so that weak points can be improved through geometry changes or material selection. A Factor of Safety (FoS) is calculated to compare the tool's strength against the expected working load; a higher FoS indicates a safer, more reliable design under real-use conditions. This section follows the same “load → stress → FoS → safety” concept.

5.4.1: Stress, Deformation, Strain Analysis and Factor of Safety Analysis for Main Body(Right & Left):

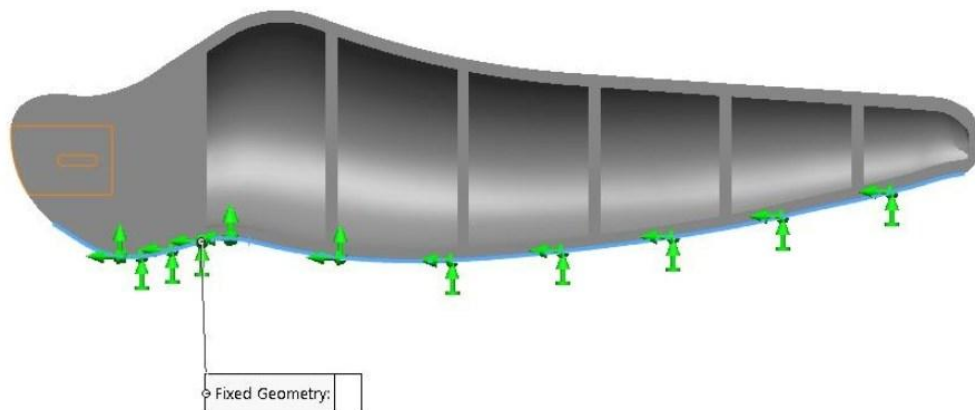


Figure 5.4.1: Fixed geometry

Applied forced on right part of the body: 20 N of force was applied on the right side of the body from the upper part.

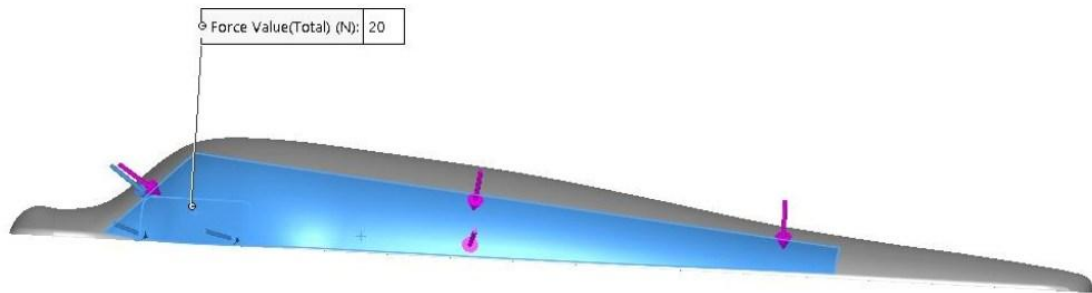


Figure 5.4.2: Applied forced on right part of the body

Table 5.4.1: Applied forced on right part of the body

Types	Force
X-Component	0 N
Y-Component	Upper portion: 20 N
Z-Component	0 N

Applied forced on Left part of the body: 20 N of force was applied on the left side of the body from the upper part along Y-axis and 10 N of force was applied along Z-axis.

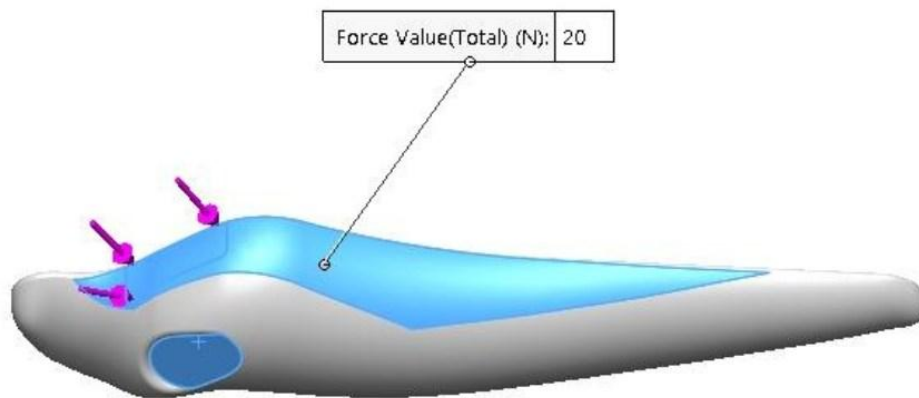


Figure 5.4.3: Applied forced on left part of the body

Table 5.4.2: Applied forced on left part of the body

Types	Force
X-Component	0 N
Y-Component	Upper portion: 20 N
Z-Component	10 N

For Right Side:

Stress analysis of the main body:

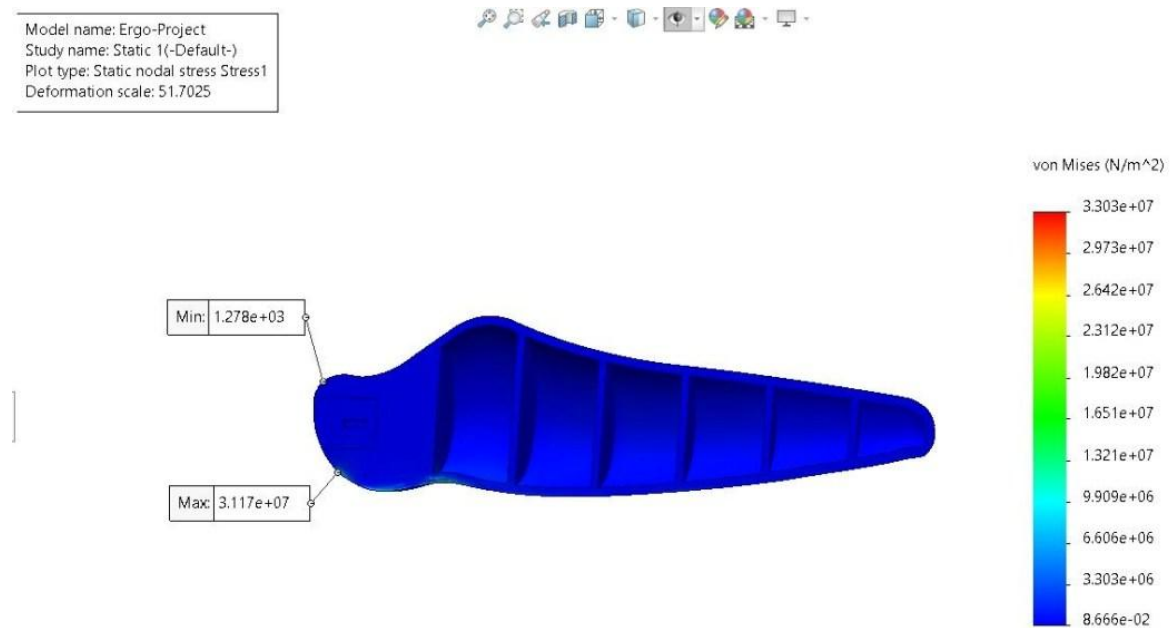


Figure 5.4.4: Stress analysis of the main body

Table 5.4.3: Stress analysis of the main body:

Stress Range Category	Lower Bound Stress (Pa)	Upper Bound Stress (Pa)
Safe Range	8.666×10^{-2}	3.303×10^6
Medium Range	3.303×10^6	1.651×10^7
Breaking Range	1.651×10^7	3.303×10^7
Maximum Allowable Stress	—	3.303×10^7

Displacement analysis of the main body:

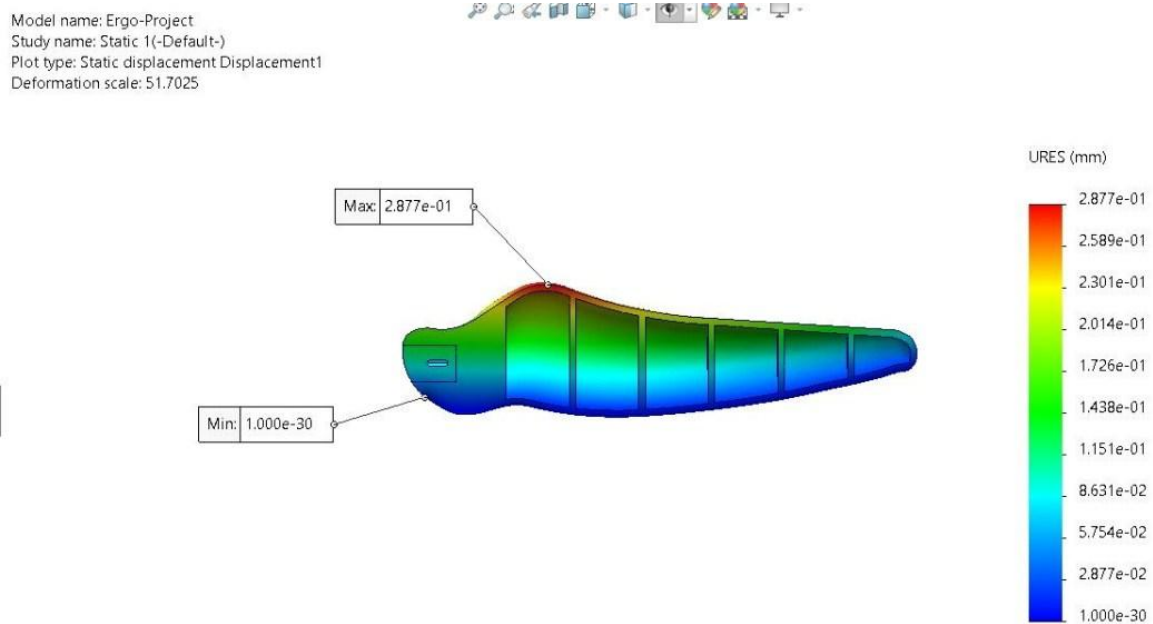


Figure 5.4.5: Displacement analysis of the main body

Table 5.4.4: Displacement analysis of the main body

Displacement Range Category	Lower Bound Displacement (mm)	Upper Bound Displacement (mm)
Safe Range	1.000×10^{-30}	2.877×10^{-2}
Medium Range	2.877×10^{-2}	1.438×10^{-1}
Breaking Range	1.438×10^{-1}	2.877×10^{-1}
Maximum Allowable Displacement	—	2.877×10^{-1}

Strain analysis of the main body

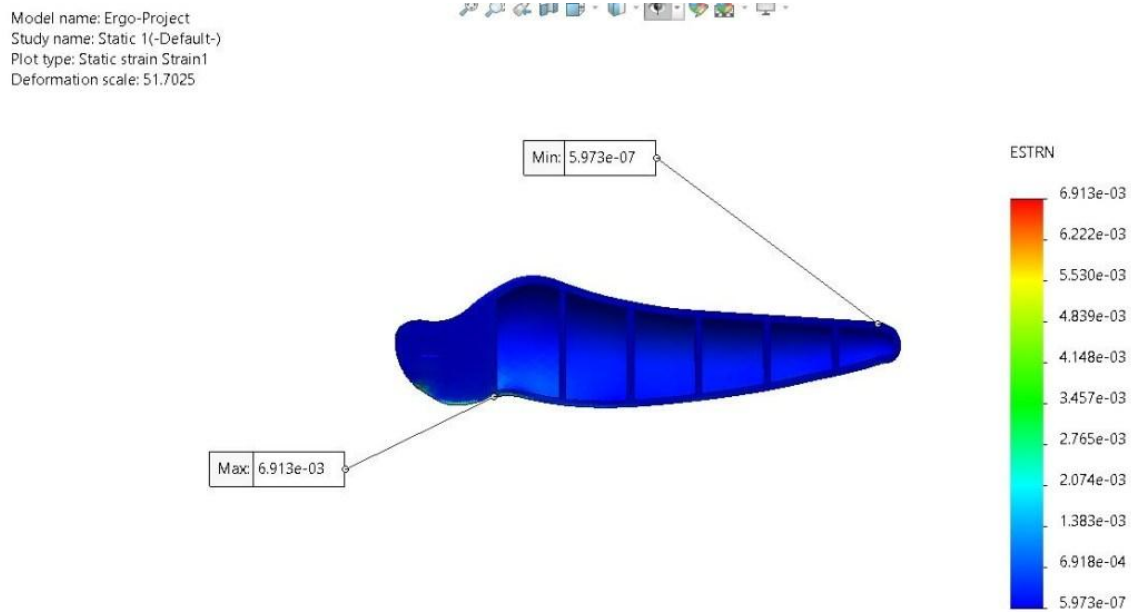


Figure 5.4.6: Strain analysis of the main body

Table 5.4.5: Strain analysis of the main body

Strain Range Category	Lower Bound Strain	Upper Bound Strain
Safe Range	5.973×10^{-7}	6.918×10^{-4}
Medium Range	6.918×10^{-4}	3.457×10^{-3}
Breaking Range	3.457×10^{-3}	6.913×10^{-3}
Maximum Allowable Strain	—	6.913×10^{-3}

Factor of Safety analysis of the main body

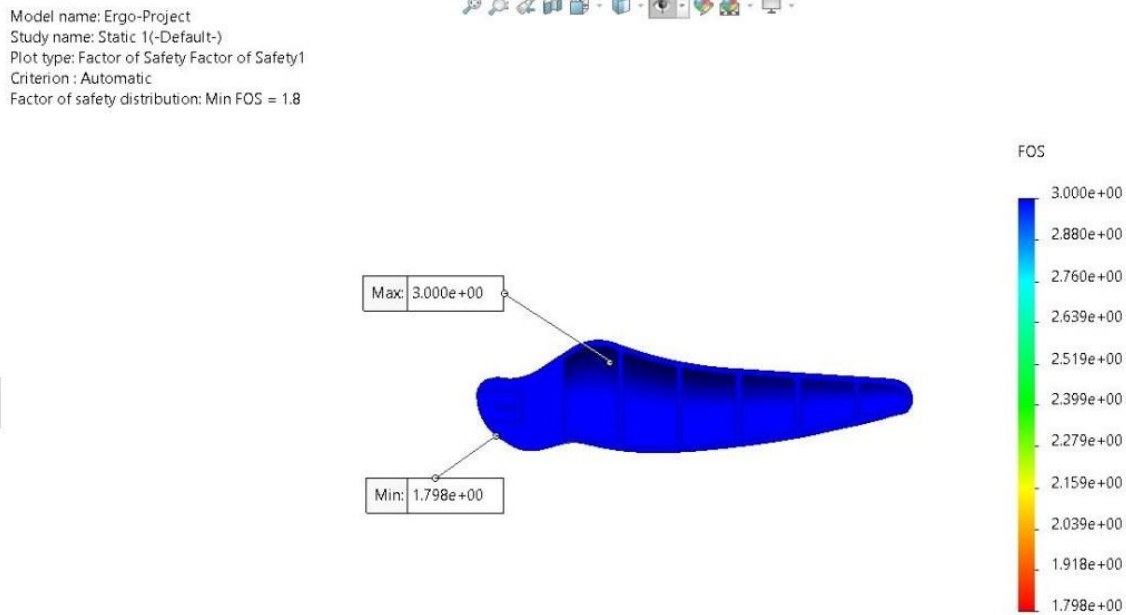


Figure 5.4.7: Factor of Safety analysis of the main body

Table 5.4.6: Factor of Safety analysis of the main body

FOS Range Category	Lower Bound FOS	Upper Bound FOS
Safe Range	2.639×10^0	3.000×10^0
Medium Range	2.279×10^0	2.639×10^0
Breaking Range	1.798×10^0	2.279×10^0
Minimum (Critical) FOS	—	1.798×10^0

For Left Side:

Stress analysis of the main body

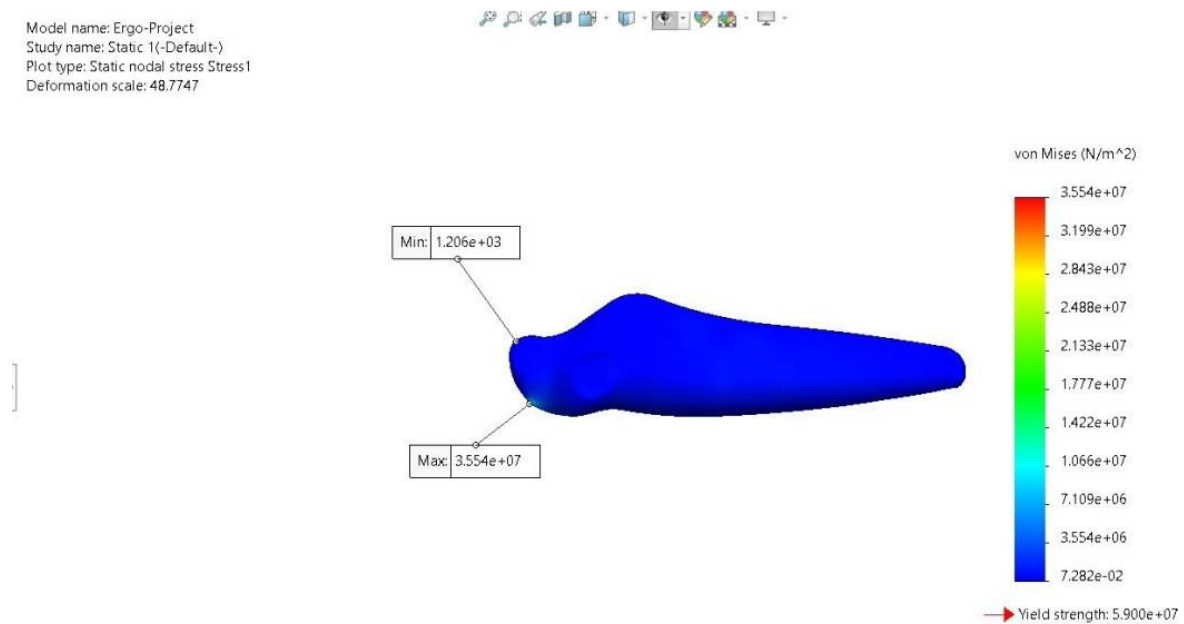


Figure 5.4.8: Stress analysis of the main body

Table 5.4.7: Stress analysis of the main body

Stress Range Category	Lower Bound Stress (Pa)	Upper Bound Stress (Pa)
Safe Range	7.282×10^{-2}	3.554×10^6
Medium Range	3.554×10^6	1.777×10^7
Breaking Range	1.777×10^7	3.554×10^7
Maximum Allowable Stress	—	3.554×10^7

Displacement analysis of the main body

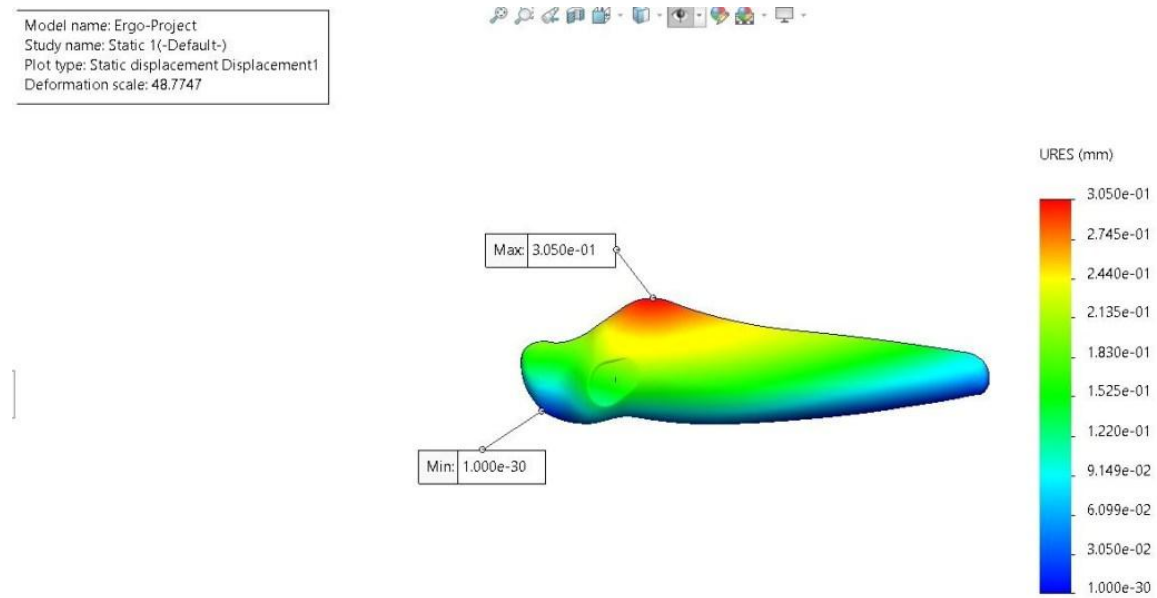


Figure 5.4.9: Displacement analysis of the main body

Table 5.4.8: Displacement analysis of the main body

Displacement Range Category	Lower Bound Displacement (mm)	Upper Bound Displacement (mm)
Safe Range	1.000×10^{-30}	3.050×10^{-2}
Medium Range	3.050×10^{-2}	1.525×10^{-1}
Breaking Range	1.525×10^{-1}	3.050×10^{-1}
Maximum Allowable Displacement	—	3.050×10^{-1}

Strain analysis of the main body

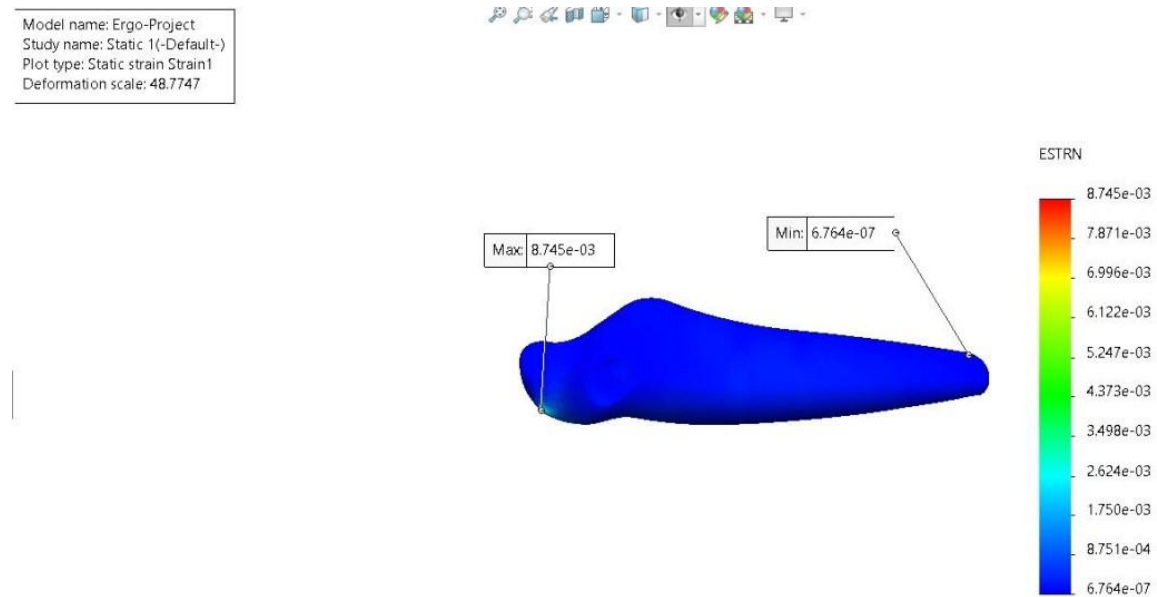


Figure 5.4.10: Strain analysis of the main body

Table 5.4.9: Strain analysis of the main body

Strain Range Category	Lower Bound Strain	Upper Bound Strain
Safe Range	6.764×10^{-7}	8.751×10^{-4}
Medium Range	8.751×10^{-4}	4.373×10^{-3}
Breaking Range	4.373×10^{-3}	8.745×10^{-3}
Maximum Allowable Strain	—	8.745×10^{-3}

Factor of Safety analysis of the main body

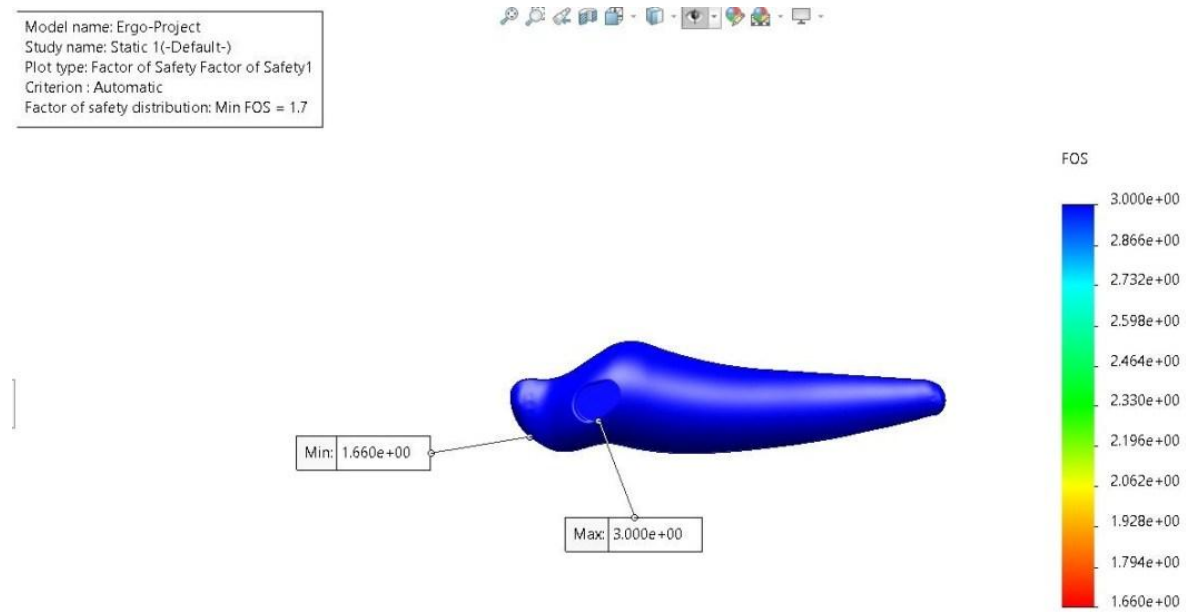


Figure 5.4.11: Factor of Safety analysis of the main body

Table 5.4.10: Factor of Safety analysis of the main body

FOS Range Category	Lower Bound FOS	Upper Bound FOS
Safe Range	2.598×10^0	3.000×10^0
Medium Range	2.196×10^0	2.598×10^0
Breaking Range	1.660×10^0	2.196×10^0
Minimum (Critical) FOS	—	1.660×10^0

5.4.2: Stress, Deformation, Strain Analysis and Factor of Safety Analysis for Blade:

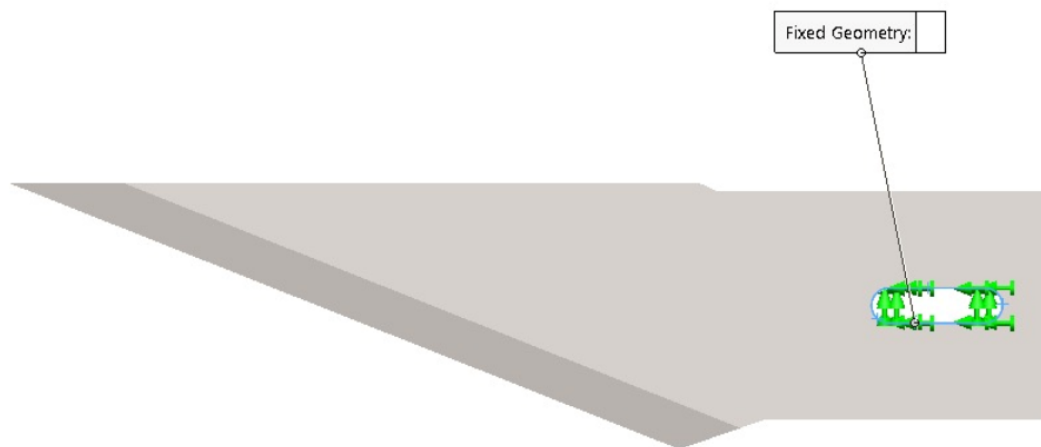


Figure 5.4.12: Fixed geometry

Applied forced on the blade: 5 N of force was applied on the edge of the blade along Y-axis

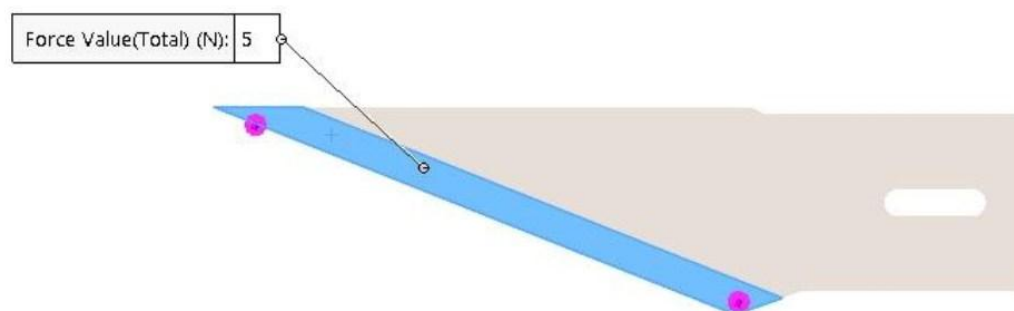


Figure 5.4.13: Applied forced on the edge of the blade

Table 5.4.11: Applied forced on the edge of the blade

Types	Force
X-Component	0 N
Y-Component	5 N (Upward)
Z-Component	0 N

Stress analysis of the blade:

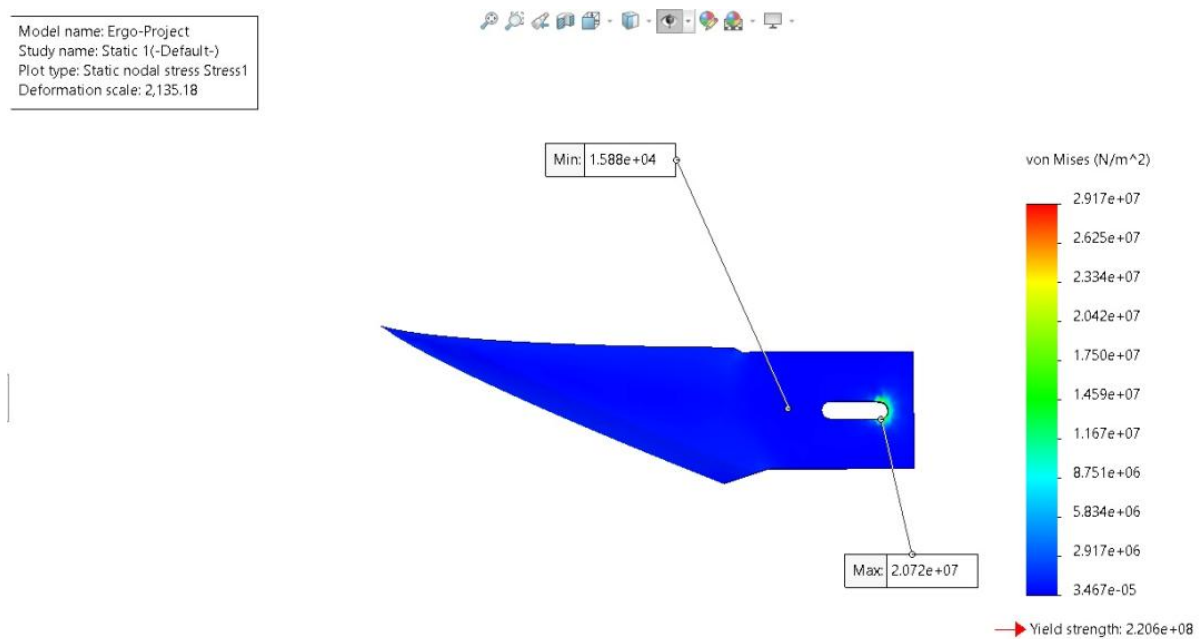


Figure 5.4.14: Stress analysis of the blade

Table 5.4.12: Stress analysis of the blade

Stress Range Category	Lower Bound Stress (Pa)	Upper Bound Stress (Pa)
Safe Range	3.467×10^{-5}	2.917×10^6
Medium Range	2.917×10^6	1.459×10^7
Breaking Range	1.459×10^7	2.917×10^7
Maximum Allowable Stress	—	2.917×10^7

Displacement analysis of the blade

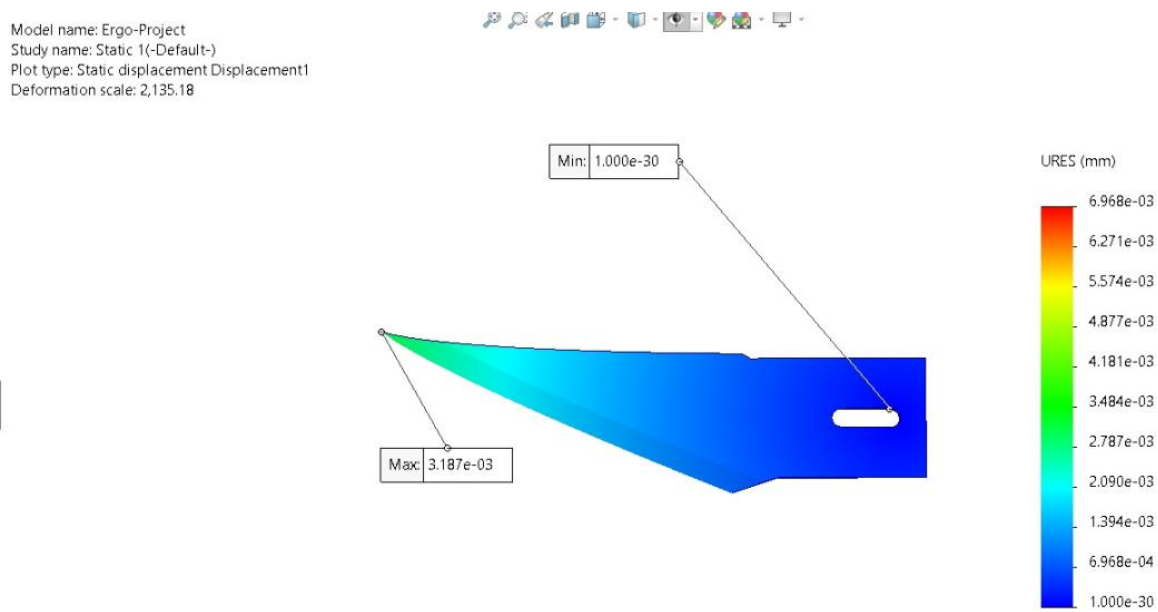


Figure 5.4.15: Displacement analysis of the blade

Table 5.4.13: Displacement analysis of the blade

Displacement Range Category	Lower Bound Displacement (mm)	Upper Bound Displacement (mm)
Safe Range	1.000×10^{-30}	6.968×10^{-4}
Medium Range	6.968×10^{-4}	3.484×10^{-3}
Breaking Range	3.484×10^{-3}	6.968×10^{-3}
Maximum Allowable Displacement	—	6.968×10^{-3}

Strain analysis of the blade

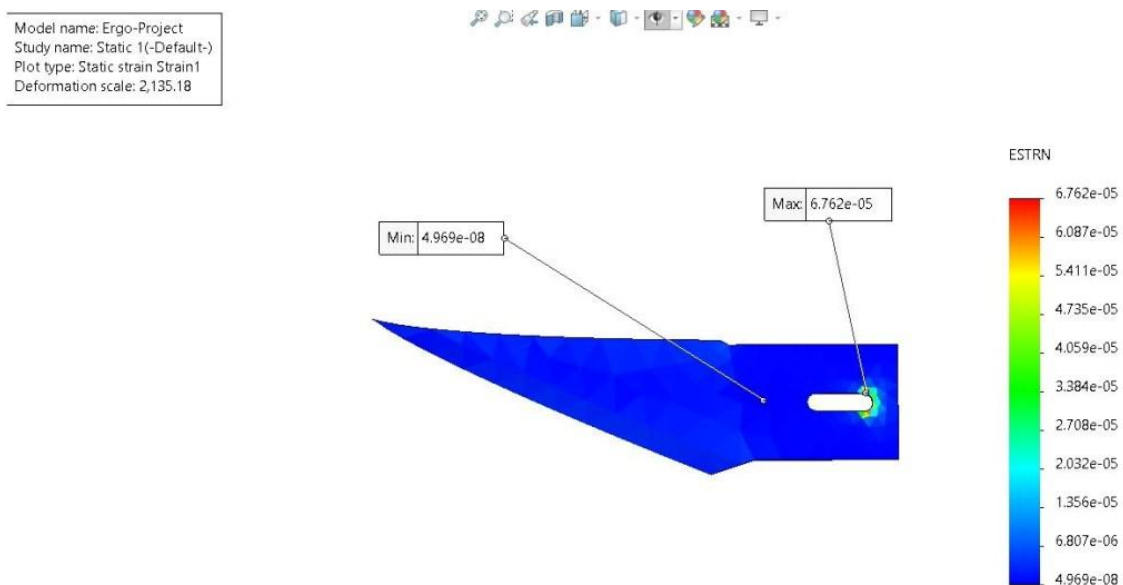


Figure 5.4.16: Strain analysis of the blade

Table 5.4.14: Strain analysis of the blade

Strain Range Category	Lower Bound Strain	Upper Bound Strain
Safe Range	4.969×10^{-8}	6.807×10^{-6}
Medium Range	6.807×10^{-6}	3.384×10^{-5}
Breaking Range	3.384×10^{-5}	6.762×10^{-5}
Maximum Allowable Strain	—	6.762×10^{-5}

Factor of safety analysis of the blade

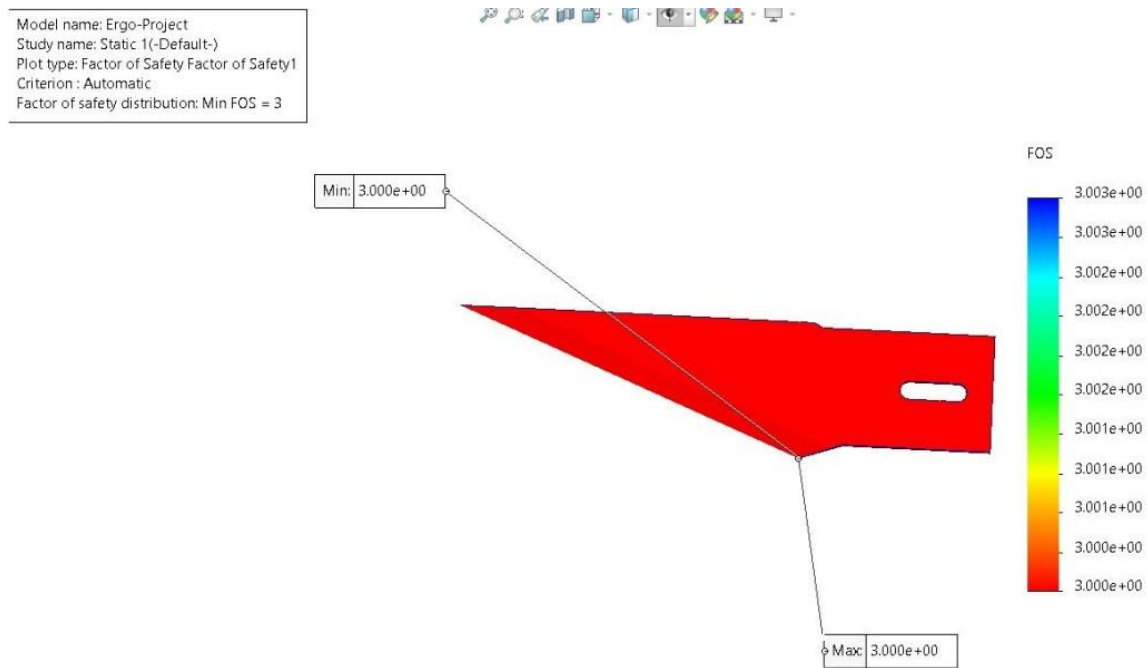


Figure 5.4.17: Factor of safety analysis of the blade

Table 5.4.15: Factor of safety analysis of the blade

FOS Range Category	Lower Bound FOS	Upper Bound FOS
Safe Range	3.002×10^0	3.003×10^0
Medium Range	3.001×10^0	3.002×10^0
Breaking Range	3.000×10^0	3.001×10^0
Minimum (Critical) FOS	—	3.000×10^0

5.4.3 Remarks for all the parts (Body & Blade):

The maximum stress obtained from the SolidWorks Simulation for the ergonomic cutting tool handle/body ($3.554\text{E}+07$ Pa) is lower than the specified yield strength of the material ($5.900\text{E}+07$ Pa). This indicates that the design remains within the elastic limit under the applied loading condition and therefore has an acceptable factor of safety for the handle/body structure.

Similarly, for the cutting blade, the maximum von Mises stress from simulation ($2.072\text{E}+07$ Pa) is significantly lower than the blade material yield strength ($2.206\text{E}+08$ Pa), which suggests a strong safety margin and conservative loading/boundary conditions in the study.

Conclusion: The same conclusion applies to the strain/deformation and factor of safety results, where the obtained maximum strain/displacement values are small and the minimum FoS values remain above 1 (e.g., 1.8 and 1.7 for the handle cases, and ~3 for the blade case), confirming that the overall design is structurally safe under the given simulation setup.

5.5 Comparison between SolidWorks Model and Manufactured Model:



Figure 5.5.1: Single best rendering photo of assembly model



Figure 5.5.2 - Manufactured Model

6.1 INTRODUCTION

Material selection is a fundamental step in engineering design, as it directly influences a product's performance, durability, safety, and cost. The success of any mechanical product depends not only on its design geometry but also on choosing materials that meet functional, mechanical, and economic requirements. A systematic material selection process helps ensure that each component performs efficiently under its expected service conditions while maintaining manufacturing feasibility and cost effectiveness.

In the case of the Ergonomic Cutting Tool, different components are subjected to different operational demands. The blade requires high yield strength, hardness, and wear resistance to maintain sharpness and withstand cutting forces. The body must provide adequate structural support while remaining lightweight to improve user handling. The grip demands materials that offer comfort, high friction, and resistance to environmental effects, whereas the cap requires sufficient impact resistance and proper fastening capability. Since these requirements vary for each part, a single material cannot satisfy all functional needs, making component-wise material selection essential.

To achieve optimal performance, both qualitative and quantitative methods were applied in this study. The Digital Logic Method was used to determine the relative importance of material properties, followed by scaling of properties and calculation of a performance index for final selection. Key criteria considered include yield strength, specific weight, cost efficiency, corrosion resistance, and availability.

Through this systematic approach, the selected materials ensure mechanical reliability, ergonomic comfort, manufacturability, and economic viability of the Ergonomic Cutting Tool. Proper material selection therefore plays a vital role in enhancing product performance, user satisfaction, and overall sustainability.

6.2 MATERIAL SELECTION PROCESS ANALYSIS

Table 6.2.1: Qualitative analysis for different parts

Parts	Selection Criteria	Available Material	Selected Material
Blade	Hardness Wear Resistance Toughness	High Carbon Steel Stainless Steel Alloy Steel	High Carbon Steel
Body	Toughness Cost Efficient Machinability Yield Strength Availability	ABS (Acrylonitrile Butadiene Styrene) Polypropylene Copolymer (PPC) PA6 (Nylon 6)	Polypropylene Copolymer (PPC)
Grip	Chemical Resistance Tactile Feel Friction	TPE (Thermoplastic Elastomer) Silicone Rubber TPU (Thermoplastic Polyurethane)	TPE (Thermoplastic Elastomer)
Cap	Impact Resistance Fastening Ability	Polypropylene (PP) ABS Plastic Nylon (PA6 / PA66)	Polypropylene (PP)

Table 6.2.2: Material Selection chart for Blade

Selection Criteria	Importance (α)	High Carbon Steel Rating	Weight	Stainless Steel Rating	Weight	Alloy Steel Rating	Weight
Hardness	0.3	5	1.5	4	1.2	4	1.2
Wear Resistance	0.25	5	1.25	4	1	4	1
Toughness	0.2	3	0.6	4	0.8	5	1
Cost Efficiency	0.15	5	0.75	3	0.45	3	0.45
Availability	0.1	5	0.5	4	0.4	4	0.4
Total	1		4.6		3.85		4.05

Table 6.2.3: Determination of Relative Importance Goals Using Digital Logic Method

Selection Criteria	Decision Number										Positive Decision	Weighting Factor (α)
	1	2	3	4	5	6	7	8	9	10		
Hardness	0	0	0	0							0	0
Wear Resistance	1				1	0	0				2	0.2
Toughness		1			0			0	0		1	0.1
Corrosion Resistance			1			1		1		0	3	0.3
Cost Efficiency				1			1		1	1	4	0.4
Total											10	$\Sigma\alpha = 1.00$

Table 6.2.4: Material Properties for Hardness, Wear Resistance, Toughness, Cost and Availability

Material	Hardness (HB)	Wear Resistance	Toughness	Cost	Availability
High Carbon Steel	600	5	3	5	5
Stainless Steel	500	4	4	3	5
Alloy Steel	550	4	5	3	3

Table 6.2.5: Scaled Values of Properties and Performance Index

Material	Scaled Properties					Performance Index (γ)
	Hardness (HB)	Wear Resistance	Toughness	Cost	Availability	
High Carbon Steel	100	100	60	100	100	92
Stainless Steel	83	80	80	75	100	82.4
Alloy Steel	92	90	100	75	80	86.6

Table 6.2.6: Material Selection chart for Body

Selection Criteria	Importance (α)	PPC Rating (1–5)	PPC Weight	ABS Rating	ABS Weight	PA6 Rating	PA6 Weight
Yield Strength	0.25	4	1	3	0.75	5	1.25
Specific Weight	0.2	5	1	4	0.8	3	0.6
Cost Efficiency	0.25	5	1.25	4	1	3	0.75
Machinability	0.15	4	0.6	4	0.6	3	0.45
Availability	0.15	5	0.75	4	0.6	3	0.45
Total	1		4.6		3.75		3.5

Table 6.2.7: Determination of Relative Importance Goals Using Digital Logic Method

Property	Decision Number										Positive Decision	Weighting Factor (α)
	1	2	3	4	5	6	7	8	9	10		
Yield Strength	1	0	1	0							2	0.2
Specific Weight	0				0	0	0				0	0.0
Cost Efficiency		1			1			1	0		3	0.3
Machinability			0			1		0		0	1	0.1
Availability				1			1		1	1	4	0.4
Total Positive Decision											10	$\Sigma\alpha = 1.0$

Table 6.2.8: Material Properties for Yield Strength and rating for Specific Weight, Cost Efficiency and Availability

Material	Yield Strength (MPa)	Specific Weight ($\frac{N}{m^3}$)	Cost Efficiency	Corrosion Resistance	Availability
PPC	30	0.9	5	5	5
ABS	40	1.04	3	4	5
PA6	75	1.14	1	4	3

Table 6.2.9: Scaled Values of Properties and Performance Index

Material	Scaled Properties (β)					Performance Index ($\gamma = \Sigma\alpha\beta$)
	Yield Strength (MPa)	Specific Weight $\frac{N}{m^3}$	Cost Efficiency	Corrosion Resistance	Availability	
PPC	40	100	100	100	100	88
ABS	53	86	75	80	100	81.1
PA6	100	79	20	80	60	58

Table 6.2.10: Material Selection chart for Grip

Selection Criteria	Importance (α)	TPE Rating	Weight	Silicone Rating	Weight	TPU Rating	Weight
Chemical Resistance	0.25	4	1	5	1.25	4	1
Tactile Feel	0.25	5	1.25	4	1	4	1
Friction	0.2	5	1	4	0.8	4	0.8
Cost Efficiency	0.15	5	0.75	3	0.45	3	0.45
Durability	0.15	4	0.6	4	0.6	5	0.75
Total	1		4.6		4.1		4

Table 6.2.11: Determination of Relative Importance Goals Using Digital Logic Method

Property	Decision Number										Positive Decision	Weighting Factor (α)
	1	2	3	4	5	6	7	8	9	10		
Tactile Feel	1	1	0	0							2	0.2
Friction	0				0	0	0				0	0
Chemical Resistance		0			1			0	0		1	0.1
Durability			1			1		1		0	3	0.3
Cost Efficiency				1			1		1	1	4	0.4
Total Positive Decision											10	$\Sigma\alpha = 1.00$

Table 6.2.12: Material Properties for Yield Strength and rating for Specific Weight, Cost Efficiency and Availability

Material	Hardness (HB)	Wear Resistance	Toughness	Cost Efficiency	Availability
TPE	70	5	4	1	5
Silicone Rubber	60	3	3	3	4
TPU	85	4	4	3	3

Table 6.2.13: Scaled Values of Properties and Performance Index

Material	Scaled Properties					Performance Index ($\gamma = \Sigma\alpha\beta$)
	Hardness (HB)	Wear Resistance	Toughness	Cost Efficiency	Availability	
TPE	82	100	80	20	100	70.4
Silicone	70	60	60	60	80	70
TPU	100	80	80	60	60	70

Table 6.2.14: Material Selection chart for Cap

Selection Criteria	Importance (α)	PP Rating	Weight	ABS Rating	Weight	Nylon Rating	Weight
Impact Resistance	0.3	4	1.2	4	1.2	5	1.5
Fastening Ability	0.25	4	1	4	1	5	1.25
Cost Efficiency	0.2	5	1	4	0.8	3	0.6
Weight	0.15	5	0.75	4	0.6	3	0.45
Availability	0.1	5	0.5	4	0.4	4	0.4
Total	1		4.45		4		4.2

Table 6.2.15: Determination of Relative Importance Goals Using Digital Logic Method

Property	Decision Number						Positive Decision	Weighting Factor (α)
	1	2	3	4	5	6		
Impact Resistance	1	0	0				1	0.16
Wear Resistance	0			0	0		0	0
Availability		1		1		1	3	0.5
Cost Efficiency			1		1	0	2	0.33
							Total Positive Decision =6	$\Sigma\alpha = 1.00$

Table 6.2.16: Material Properties for Yield Strength and rating for Specific Weight, Cost Efficiency and Availability

Material	Impact Strength (kJ/m ²)	Density (g/cm ³)	Moldability	Cost Efficiency
Polypropylene (PP)	40	0.9	5	5
ABS	45	1.04	4	3
Nylon (PA6)	35	1.14	3	1

Table 6.2.17: Scaled Values of Properties and Performance Index

Material	Scaled Properties				Performance Index ($\gamma = \sum \alpha\beta$)
	Impact Strength (kJ/m ²)	Density (g/cm ³)	Moldability	Cost Efficiency	
PP	88	100	100	100	97.1
ABS	100	86	80	60	75.8
Nylon	78	79	60	20	49

6.3 Manufacturing Process Selection

The selection of a manufacturing process is a critical decision in product development that involves identifying the most suitable method for producing a component. This process requires a careful evaluation of various techniques to determine which one best aligns with the product's design requirements and the company's strategic goals. Key factors influencing this decision include unit cost, production speed, surface quality, and material availability. While this selection process can be intricate, it is essential for ensuring that the final product is manufactured efficiently, cost-effectively, and to the highest standard of quality. The chosen process significantly dictates the commercial viability and mechanical integrity of the final product.

For this project, we will adopt a hybrid strategy. Standardized components, such as fastening screws and packaging materials, will be purchased from external suppliers. However, the core

components of the ergonomic knife will be manufactured in factory, transforming raw materials into the specified shapes and sizes. Specifically, we need to manufacture the ergonomic body from high-impact polymer and the blade from hardened steel.

There are two main functional parts of this product:

- Ergonomic Hand (Body part)
- Knife Blade

For each part, we have considered specific manufacturing operations:

- Part Shaping
- Cutting

Table 6.3.1: Manufacturing process analysis for different parts

Process	Part	Selection Criteria	Available Process	Selected Process
Part Shaping	-Ergonomic Handle	-Cost (Unit Price) -Availability -Production Speed -Surface Finish	-Injection Molding -3D Printing -Vacuum Casting	Injection Molding
Cutting	-Knife Blade	-Cost (Unit Price) -Availability -Operation Time -Quality	-Stamping (Fine Blanking) -Laser Beam Cutting -Abrasive Water Jet (AWJM)	Stamping (Fine Blanking)

Table 6.3.2: Manufacturing Process Selection Chart for part shaping

Selection Criteria	Importance	Injection Rating	Injection Weight	3D Print Rating	3D Print Weight	Vacuum Rating	Vacuum Weight
Cost	30%	5	1.5	2	0.6	3	0.9
Availability	10%	3	0.3	5	0.5	4	0.4
Speed	30%	5	1.5	1	0.3	3	0.9
Finish	30%	5	1.5	2	0.6	4	1.2
Total	100%		4.8		2.0		3.4

Result: As the weight is greatest (4.8) for Injection Molding, we should use this process for the part shaping operation.

Table 6.3.3: Determination of Relative Importance Goals Using Digital Logic Method

Selection Criteria	Decision Number						Positive Decisions	Weighting Factor (α)
	1	2	3	4	5	6		
Cost	0	0	1				1	0.17
Availability	1			0	1		2	0.33
Production Speed		1		1		0	2	0.33
Surface Finish			0		0	1	1	0.17
						Total	6	$\sum \alpha = 1.0$

Table 6.3.4: Rating for Cost, Availability, Production Speed and Surface Finish

Machining Process	Cost (Unit Price)	Availability	Operation Time	Surface Finish
Injection Molding	1	3	1	5
3D Printing	5	5	5	2
Vacuum Molding	3	4	2	4

Table 6.3.5: Scaled Value of Properties and Performance Index

Machining Process	Scaled Properties (β)				Performance Index (γ)
	Cost	Availability	Production Time	Surface Finish	
Injection Molding	100	60	100	100	86.8
3D Printing	20	100	20	40	49.8
Vacuum Molding	33	80	50	80	62.1

Result: The process performance index is greatest (86.8) for Injection Molding. So, we should use this process for the part shaping operation.

Formula Used:

1. For goals: Availability, Surface finish-

$$\text{Scaled Property, } \beta = \frac{\text{Numerical value of property}}{\text{Maximum Value}} \times 100$$

2. For goals: Unit Cost, Production Time,

$$\text{Scaled Property, } \beta = \frac{\text{Minimum value in list}}{\text{Numerical value in the list}} \times 100$$

3. The material performance index (γ) is calculated -

$$\gamma = \sum \alpha \beta$$

Table 6.3.6: Manufacturing Process Selection Chart for Cutting

Selection Criteria	Importance	Stamping Rating	Stamping Weight	Laser Rating	Laser Weight	AWJM Rating	AWJM Weight
Cost	40%	5	2.0	3	1.2	2	0.8
Availability	30%	3	0.9	5	1.5	4	1.2
Less operation time	10%	5	0.5	3	0.3	2	0.2
Quality	20%	5	1.0	3	0.6	4	0.8
Total	100%		4.4		3.6		3.0

Result: As the weight is greatest (4.4) for Stamping (Fine Blanking), so we should use Stamping for the cutting operation. We can also use the Digital Logic Method.

Table 6.3.7: Determination of Relative Importance Goals Using Digital Logic Method

Selection Criteria	Decision Number						Positive Decisions	Weighting Factor (α)
	1	2	3	4	5	6		
Cost	0	1	0				1	0.17
Availability	1			1	0		2	0.33
Operation Time		0		0		1	1	0.17
Quality			1		1	0	2	0.33
						Total	6	$\sum \alpha = 1.0$

Table 6.3.8: Rating for Cost, Availability, Operation Time and Quality

Machining Process	Cost	Availability	Operation Time	Quality
Stamping (Fine Blanking)	1	3	1	5
Laser Beam Cutting	3	5	3	3
AWJM	4	4	4	4

Table 6.3.9: Scaled Value of Properties and Performance Index

Machining Process	Scaled Properties (β)				Performance Index (γ)
Stamping (Fine Blanking)	100	60	100	100	86.8
Laser Beam Cutting	33	100	33	60	64.0
AWJM	25	80	25	80	61.3

Formula Used(Material & Manufacturing Process Selection):

1. For goals: Availability, Surface finish-

$$\text{Scaled Property, } \beta = \frac{\text{Numerical value of property}}{\text{Maximum Value}} \times 100$$

2. For goals: Unit Cost, Production Time,

$$\text{Scaled Property, } \beta = \frac{\text{Minimum value in list}}{\text{Numerical value in the list}} \times 100$$

3. The material performance index (γ) is calculated -

$$\gamma = \sum \alpha \beta$$

6.4 Conclusion

For manufacturing an ergonomic knife, it is very important to identify the most important properties and selection criteria such as cost, availability, production speed, surface finish, and quality. Proper evaluation of these properties helps in giving the right priority to each factor and ensures the correct selection of materials and manufacturing processes. It is also necessary to compare different available processes carefully so that the most suitable and efficient process can be chosen without making any wrong decision.

In this project, different manufacturing processes were analysed for both the ergonomic handle and the knife blade. Based on performance index and weighted analysis, Injection Molding was selected for the ergonomic handle because it provides better surface finish, faster production, and lower cost for mass production. Similarly, stamping (Fine Blanking) was selected for the knife blade because it ensures high cutting quality, accuracy, and efficient operation. These selections help to achieve both functional performance and good product quality.

Therefore, proper identification of important properties and careful comparison of manufacturing processes plays a key role in successful product development. It helps in selecting the most suitable manufacturing methods, ensuring efficient production, improved product performance, and overall commercial success of the ergonomic knife.

Chapter 7: Design of Experiment (DOE)

Design Summary

Taguchi Array L9(3³)

Factors: 3

Runs: 9

Columns of L9(3⁴) array: 1 2 3

Parameter Table:

Layer Height (mm)	Infill Density (%)	Build Orientation (°)	Max Bending Load (N)	Dimensional Error (mm)
0.12	20	0	410	0.18
0.2	40	45	395	0.2
0.28	60	90	330	0.24
0.12	20	45	340	0.26
0.2	40	90	285	0.3
0.28	60	0	440	0.25
0.12	20	90	230	0.42
0.2	40	0	360	0.34
0.28	60	45	370	0.36

Taguchi Analysis: Max Bending Load (N), Dimensional Error (mm) versus Layer Height (mm), Infill Density (%), Build Orientation (°)

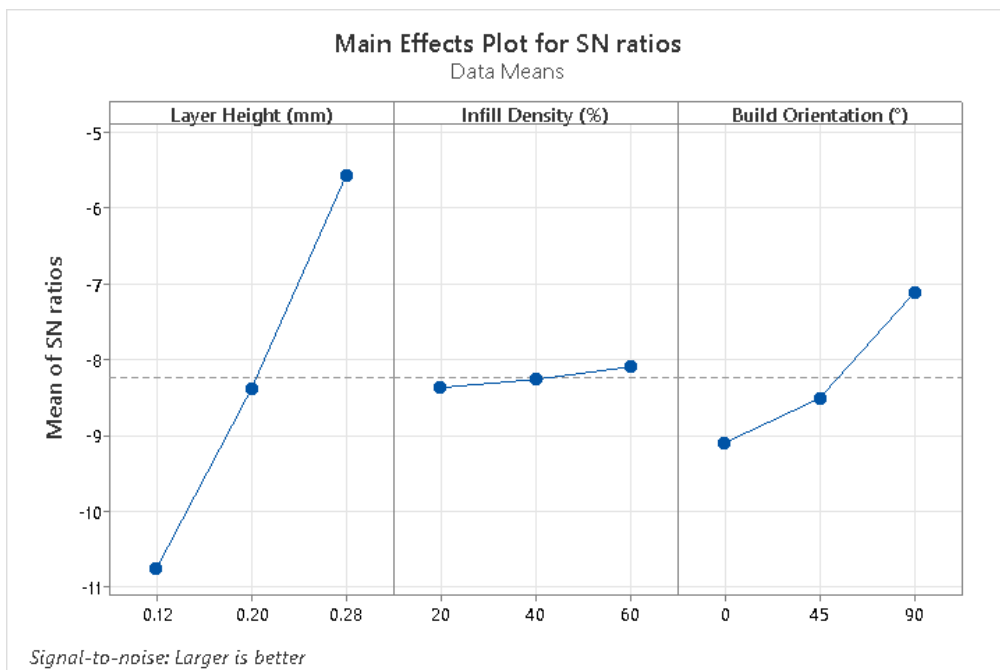
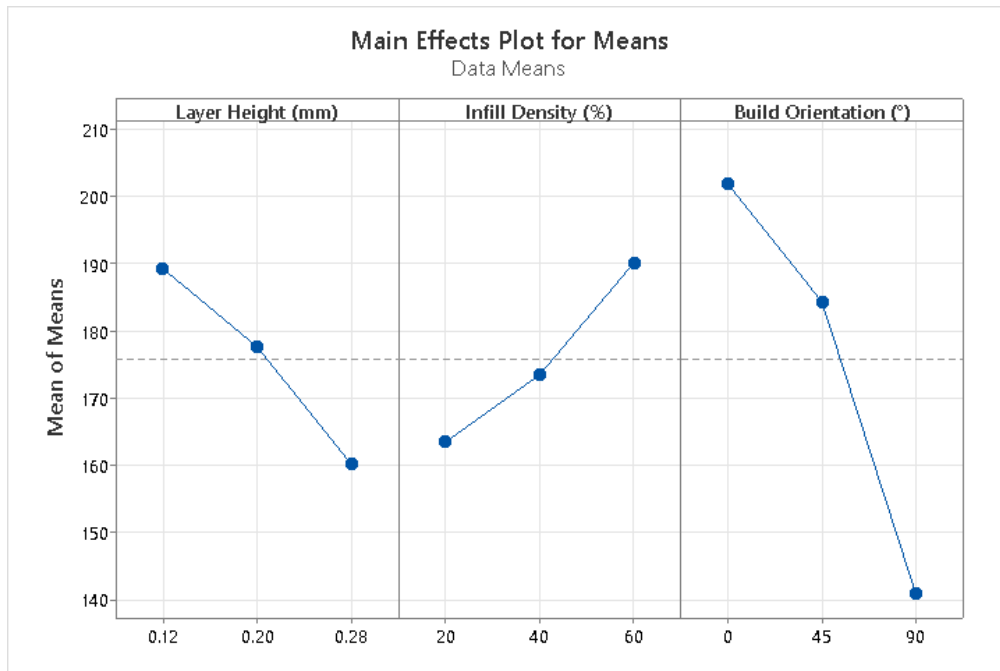
Response Table for Signal to Noise Ratios

(Larger is better)

Level	Layer Height (mm)	Infill Density (%)	Build Orientation (°)
1	-10.746	-8.366	-9.092
2	-8.389	-8.259	-8.508
3	-5.583	-8.093	-7.119
Delta	5.163	0.273	1.973
Rank	1	3	2

Response Table for Means

Level	Layer Height (mm)	Infill Density (%)	Build Orientation (°)
1	189.3	163.5	201.8
2	177.6	173.5	184.3
3	160.2	190.1	141.0
Delta	29.1	26.7	60.8
Rank	2	3	1



Regression Analysis: Max Bending Load (N) versus Layer Height (mm), Infill Density (%), Build Orientation (°)

Regression Equation

$$\text{Max Bending Load (N)} = 431.5 - 364.6 \text{ Layer Height (mm)} + 1.333 \text{ Infill Density (\%)} - 1.352 \text{ Build Orientation (°)}$$

Coefficients

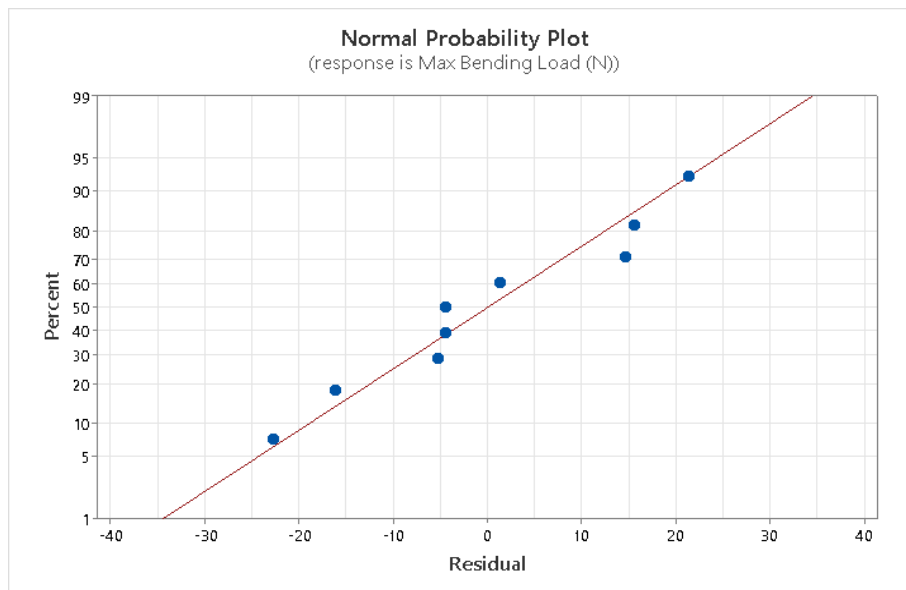
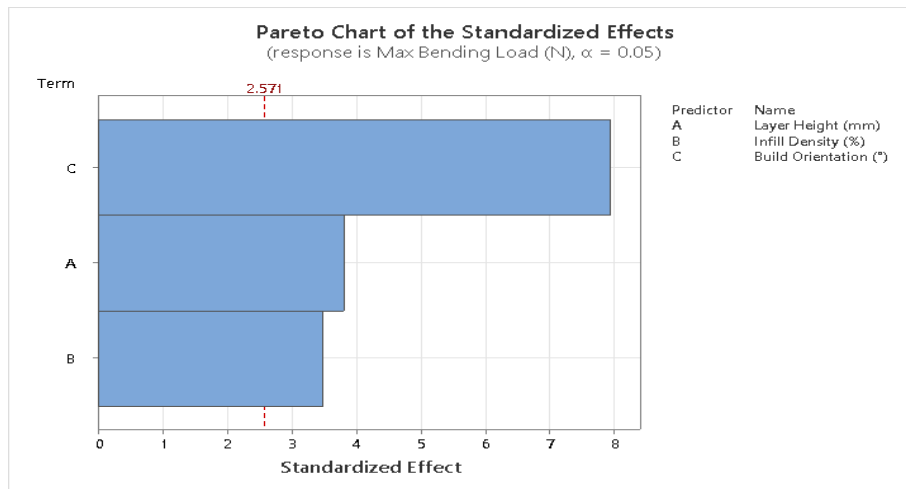
Term	Coef	SE Coef	T-Value	P-Value	VIF
Constant	431.5	26.5	16.30	0.000	
Layer Height (mm)	-364.6	95.8	-3.80	0.013	1.00
Infill Density (%)	1.333	0.383	3.48	0.018	1.00
Build Orientation (°)	-1.352	0.170	-7.93	0.001	1.00

Model Summary

S	R-sq	R-sq(adj)	R-sq(pred)
18.7824	94.71%	91.53%	81.76%

Analysis of Variance

Source	DF	Adj SS	Adj MS	F-Value	P-Value
Regression	3	31575	10525.0	29.83	0.001
Layer Height (mm)	1	5104	5104.2	14.47	0.013
Infill Density (%)	1	4267	4266.7	12.09	0.018
Build Orientation (°)	1	22204	22204.2	62.94	0.001
Error	5	1764	352.8		
Total	8	33339			



Regression Analysis: Dimensional Error (mm) versus Layer Height (mm), Infill Density (%), Build Orientation (°)

Regression Equation

$$\text{Dimensional Error (mm)} = 0.0467 + 1.0417 \text{ Layer Height (mm)} - 0.000083 \text{ Infill Density (\%)} + 0.000704 \text{ Build Orientation (°)}$$

Coefficients

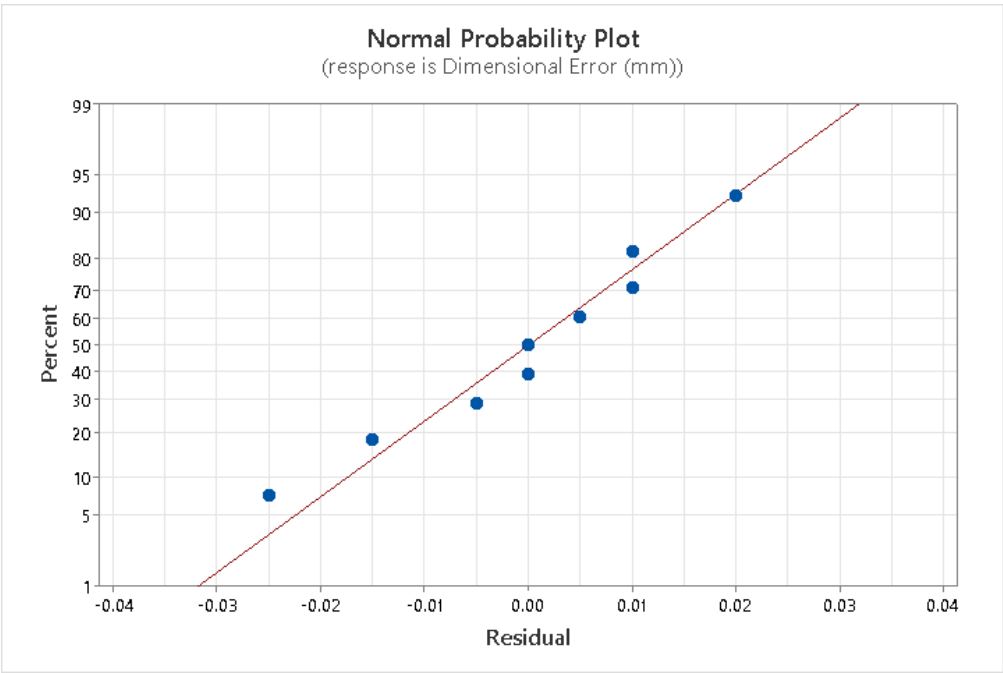
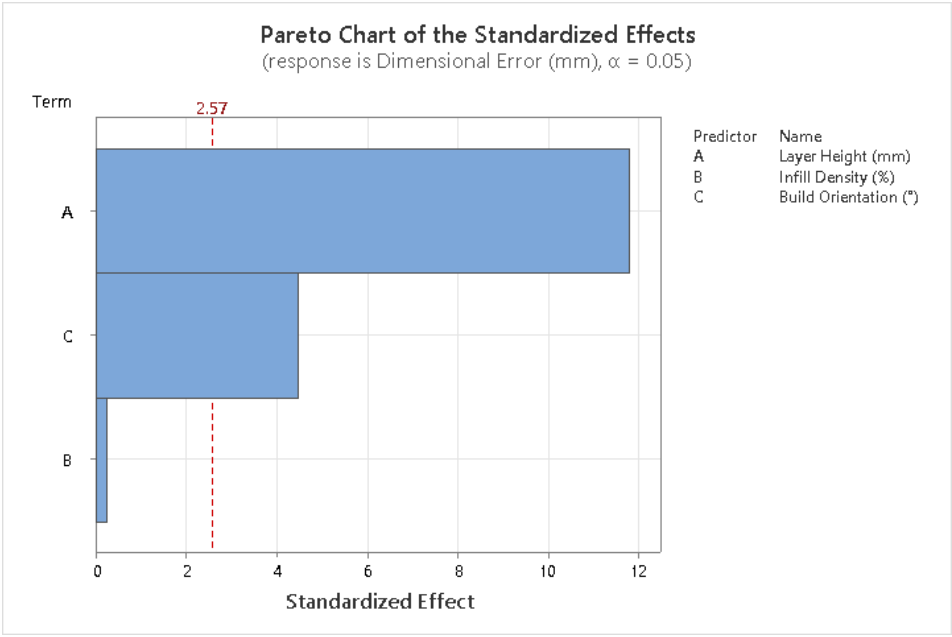
Term	Coef	SE Coef	T-Value	P-Value	VIF
Constant	0.0467	0.0244	1.91	0.114	
Layer Height (mm)	1.0417	0.0884	11.79	0.000	1.00
Infill Density (%)	-0.000083	0.000354	-0.24	0.823	1.00
Build Orientation (°)	0.000704	0.000157	4.48	0.007	1.00

Model Summary

S	R-sq	R-sq(adj)	R-sq(pred)
0.0173205	96.95%	95.12%	88.29%

Analysis of Variance

Source	DF	Adj SS	Adj MS	F-Value	P-Value
Regression	3	0.047700	0.015900	53.00	0.000
Layer Height (mm)	1	0.041667	0.041667	138.89	0.000
Infill Density (%)	1	0.000017	0.000017	0.06	0.823
Build Orientation (°)	1	0.006017	0.006017	20.06	0.007
Error	5	0.001500	0.000300		
Total	8	0.049200			



Chapter 08

Cost Analysis

8.1 INTRODUCTION

Cost analysis is a fundamental part of product design and development. It helps determine the economic feasibility, profitability, and sustainability of a product throughout its entire life cycle. According to the cost analysis framework, costs can generally be classified into fixed costs and variable costs. Proper identification and evaluation of these costs are essential before launching a product into the market.

The product considered in this report is an Ergonomic Cutting Tool, designed to provide improved comfort, reduced hand fatigue, and enhanced safety during cutting operations. The tool is intended for use in industrial, workshop, and household environments where repetitive cutting tasks are common. The ergonomic design focuses on better grip support, optimized handle shape, and efficient blade positioning to improve user performance and productivity.

This report presents a detailed cost analysis of the Ergonomic Cutting Tool by examining its cost structure throughout the product life cycle. The analysis includes research and development (R&D) costs, facilities expenses, production costs (materials, labour, overhead), marketing, distribution, customer service, and other additional overhead costs as outlined in the lifecycle cost structure. Furthermore, fixed and variable cost components will be identified to support break-even and profitability evaluation.

The objective of this report is to estimate the total cost per unit, evaluate overall investment requirements, and assess the financial viability of mass production of the Ergonomic Cutting Tool.

8.2 BILL OF MATERIAL

Sl. No.	Component Name	Material	Quantity	Unitary Cost (BDT)	Total Unitary Cost (BDT)	% of Material Cost
1	Cutting Blade	High Carbon Steel	1	45	45	28%
2	Right Side Body	Polypropylene Copolymer (PPC)	1	30	30	19%
3	Left Side Body	Polypropylene Copolymer (PPC)	1	30	30	19%
4	Grip	TPE (Thermoplastic Elastomer)	2	20	40	25%
5	Cap	Polypropylene (PP)	1	15	15	9%
					Total Unitary Cost = 160 BDT	100%

8.3 LIFECYCLE COST

Category	Cost Item	Year 1 (BDT)	Year 2 (BDT)	Year 3 (BDT)	Year 3 (BDT)	Lifecycle Fixed Cost (BDT)	Variable Cost per Unit (BDT)
R&D	Product Design	7000	–	–		7000	–
	Design Improvement	–	3000	2000		5000	–
	Total R&D	7000	3000	2000		12000	–
Facilities Expenses	Tooling & Setup	8000	–	–		8000	–
	Depreciation	4000	4000	4000	4000	16000	–
	Total Facilities	12000	4000	4000	4000	24000	–
Production	BOM (Materials)	–	–	–		–	160
	Labour Cost	–	–	–		–	80
	Maintenance	10000	10000	10000	10000	40000	6
	Utilities	12000	12000	12000	12000	48000	8
	Insurance	3000	3000	3000	3000	12000	–

	Total Production Cost	25000	25000	25000	25000	100000	254
Marketing	Advertisement	12000	10000	8000	8000	38000	—
	Promotion per unit	—	—	—		—	12
	Total Marketing Cost	12000	10000	8000	8000	38000	12
Distribution	Logistics Management	10000	10000	10000	10000	40000	—
	Shipping per unit	—	—	—		—	18
	Total Distribution Cost	10000	10000	10000	10000	40000	18
Customer Service	Repair & Support	6000	6000	6000	6000	24000	—
	Service per unit	—	—	—		—	8
	Total Customer Service Cost	6000	6000	6000	6000	24000	8
Additional Overheads	Administration	6000	6000	6000	6000	24000	—
	Software & Licenses	2000	2000	2000	2000	8000	—
	Total Overheads	8000	8000	8000	8000	32000	—
End of Lifecycle	Waste Management	—	—	—	3000	3000	—
	Total EOL Costs				3000	3000	
	Total Accumulated Costs	80000	66000	63000	64000	273000	292

8.4 SALES SCENERIO

Year	Sales
1	5000
2	10000
3	30000
4	20000

8.5 BREAK EVEN ANALYSIS

Break-even analysis is a financial and managerial tool used to determine the level of sales at which total revenue equals total cost. At this point, the organization neither earns a profit nor experiences a loss, meaning all expenses involved in the production and operation process are fully recovered. Break-even analysis helps businesses understand the minimum number of units that must be sold to cover all costs and begin generating profit beyond that level. It is an important method for planning production, setting selling prices, and evaluating the financial feasibility of a project.

8.5.1 COST CALCULATION

Sales Price Per Unit, $S = 450$ BDT

Number of Units Produced in Year 1, $Q = 5000$ Units

Total Revenue,

$$TR = S \times Q = 450 \times 5000 = 2,250,000 \text{ BDT}$$

Variable Cost Per Unit,

$$V = 292 \text{ BDT}$$

Year 1 Fixed Cost,

$$F = 80,000 \text{ BDT}$$

Break-Even Point,

$$BEP_x = F / (S - V)$$

$$= 80,000 / (450 - 292)$$

$$= 80,000 / 158$$

$$= 506.33 \approx 507 \text{ units}$$

Break-Even Sales Value,

$$BEP(BDT) = BEP_x \times S$$

$$= 507 \times 450$$

$$= 228,150 \text{ BDT}$$

So, 507 products should be sold to reach the break-even point.

Total Cost,

Total Cost = Fixed Cost + (Variable Cost Per Unit × Number of Units Produced Per Year)

= 80,000 + (292 × 5000)

= 80,000 + 1,460,000

= 1,540,000 BDT

Therefore,

Profit = Total Revenue - Total Cost

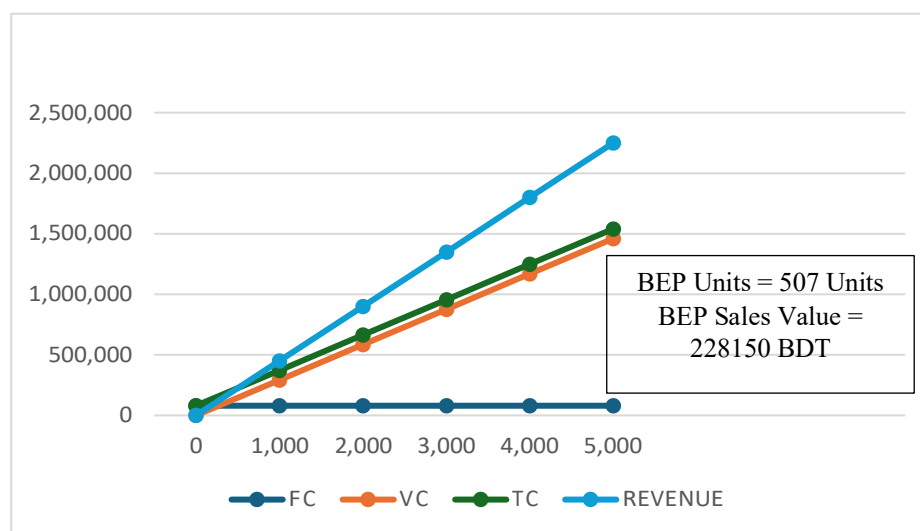
= 2,250,000 - 1,540,000

= 710,000 BDT

TABLE 8.5.2: BREAK-EVEN DATA TABLE

Unit Sold	Fixed Cost FC (BDT)	Variable Cost VC (BDT)	Total Cost (BDT)	Sales Revenue (BDT)
0	80,000	0	80,000	0
1,000	80,000	292,000	372,000	450,000
2,000	80,000	584,000	664,000	900,000
3,000	80,000	876,000	956,000	1,350,000
4,000	80,000	1,168,000	1,248,000	1,800,000
5,000	80,000	1,460,000	1,540,000	2,250,000

TABLE 8.5.3: BREAK-EVEN CHART



8.6 Conclusion

The cost analysis of the proposed cutting tool indicates that the product is financially feasible and suitable for production. Based on the bill of materials and lifecycle cost estimation, the unit cost remains within an acceptable range compared to the selected selling price of 450 BDT. The break-even analysis shows that the project can recover its fixed cost after selling about 507 units in Year 1, which is much lower than the projected sales volume. The sales scenario over four years also suggests steady market demand and good profit potential. Therefore, based on the estimated data, the product can be considered a viable and profitable solution for future production and market expansion.

Chapter 09

Conclusion

This project focused on designing an ergonomic cutting tool using a systematic product design approach. Through market analysis and QFD, key user needs were identified and translated into design requirements. FSD and design analysis helped develop an efficient and user-friendly structure, while DOE supported the optimization of key parameters. Appropriate material selection and cost analysis ensured durability and economic feasibility.

The proposed design improves user comfort, control, and overall cutting efficiency compared to conventional tools. However, the project has some limitations, such as limited real-life testing and possible variation in user comfort. Future work can focus on further testing, design refinement, and optimization for large-scale production.